Dynamical Systems

Part II

Recorded from the lecturer in 2026-2027.

Dynamical systems – mathematical entities described by a set of equations for a set of variables w.r.t. a time-like variable.

State space or phase space.

∗ Continuous time ⇒ set of ODEs

Lotka–Volterra for rabbits (r) & sheep (s)

$$\dot{r} = r(a - br - cs), \qquad \dot{s} = s(d - er - fs), \qquad a, b, \ldots, f > 0.$$

$$\text{State space} \qquad [0, \infty) \times [0, \infty).$$

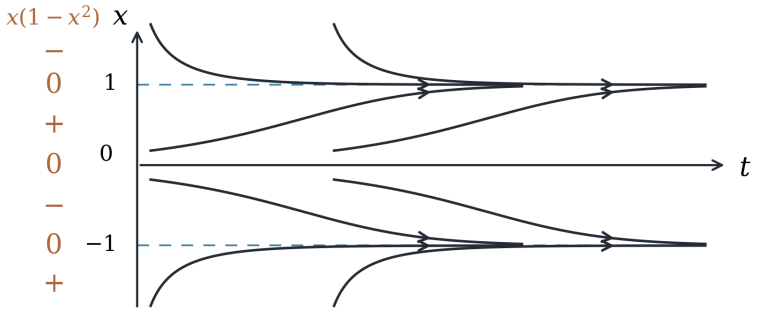
e.g. Logistic map $\qquad x_{n+1} = \mu x_n (1 - x_n), \quad 0 \le \mu \le 1$

Questions to address

- ∗ Long-term behavior of the system?
- ∗ Are there periodic solutions
- ∗ Stability of solutions, chaos
- ∗ Effect of system parameter variation

Example 1D system $\qquad \dot{x} = x(1 - x^2)$

$$\text{Solution} \qquad x(t) = \pm \left(1 + 2e^{2(t_0 - t)}\right)^{-1/2}$$



as $t \to \infty$

$x(0) > 0 \Rightarrow x(t) \to 1$

$x(0) < 0 \Rightarrow x(t) \to -1$

$x(0) = 0 \Rightarrow x(t) = 0$

Example 2D, Lotka–Volterra

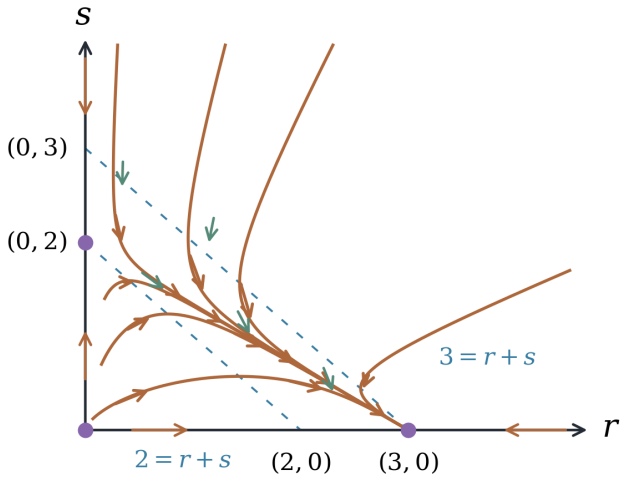
$$\dot{r} = r(3 - r - s)$$
$$\dot{s} = s(2 - r - s)$$

Can get $r(s) \Rightarrow t(s)$

∗ Find fixed points

$$\dot{r}, \dot{s} = 0 : \quad (0,0), (3,0), (0,2)$$

∗ Look at $\dot{r}, \dot{s}$ in state space



$$\text{As } t \to \infty \qquad (r,s) \to \begin{cases} (3,0), & r > 0, \\ (0,2), & r = 0,\ s > 0, \\ (0,0), & r = s = 0. \end{cases}$$

Example predator-prey

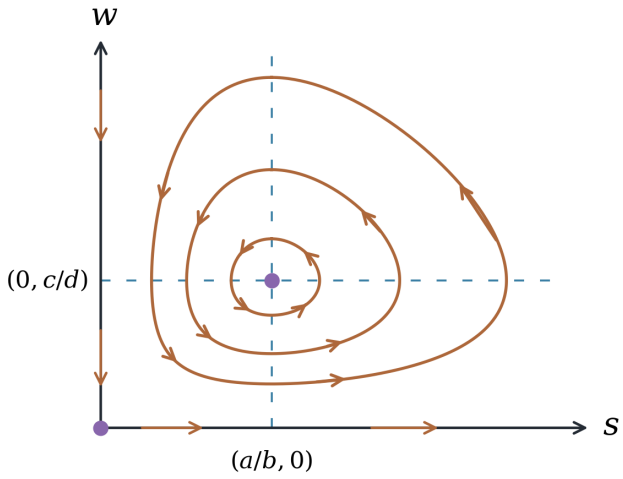
sheep (s) & wolves (w)

$$\dot{w} = w(-a + bs)$$
$$\dot{s} = s(c - dw)$$

$$a, \ldots, d > 0$$

Fixed points

$$(0,0), \quad (a/b, c/d)$$



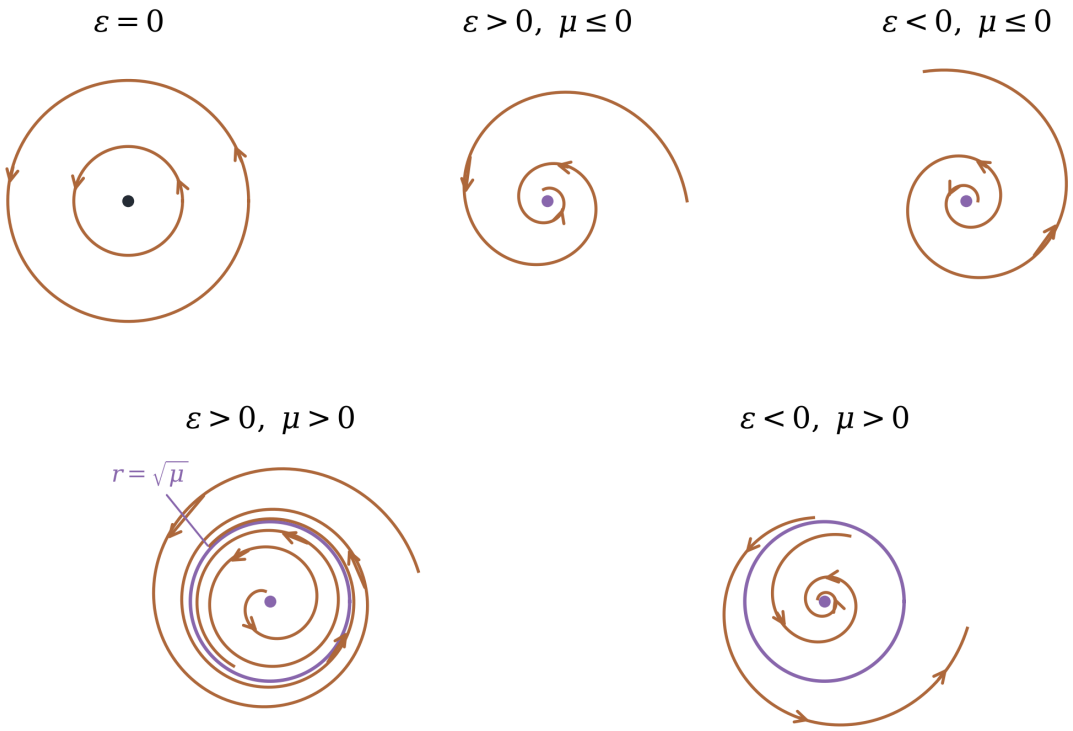
$$(s,w) \to \begin{cases} \text{periodic around } (a/b, c/d), & s > 0,\ w > 0, \\ (0,0), & s = 0,\ w \geq 0, \\ (\infty, 0), & s > 0,\ w = 0 \end{cases} \qquad \text{as } t \to \infty$$

Example

$$\dot{x} = -y + \varepsilon x(\mu - x^2 - y^2)$$
$$\dot{y} = \quad x + \varepsilon y(\mu - x^2 - y^2)$$

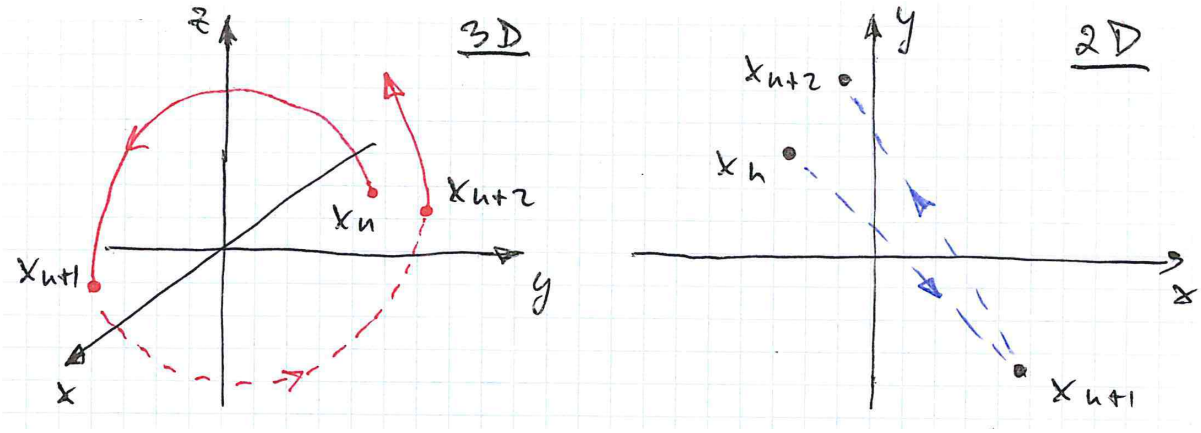
Switch to polars:

$$r = \sqrt{x^2 + y^2}, \qquad \theta = \operatorname{atan} \frac{y}{x},$$
$$\dot{r} = \frac{x\dot{x} + y\dot{y}}{r}, \qquad \dot{\theta} = \frac{x\dot{y} - y\dot{x}}{r^2} \quad \Rightarrow$$
$$\underline{\dot{r} = \varepsilon r(\mu - r^2)}, \qquad \underline{\dot{\theta} = 1}$$



End of lecture 1

Poincare map in 2D for a 3D dynamical system:



Can also consider maps as discretized ODEs.

Plan for the course

(3)	§1 :	Basic definitions	ODEs
(3)	§2 :	Fixed points	
(2)	§3 :	Stability	
(5)	§4 :	Periodic orbits	
(5)	§5 :	Bifurcations	
(3)	§6 :	Maps	Maps
(3)	§7 :	Chaos	

§1. Definitions

(1.1) Define a State Space (Phase Space) $E \subseteq \mathbb{R}^n$. The state of the system is described by $\bar{x} \in E$. The state evolves in time according to an ODE of the form

$$\dot{\bar{x}} = \frac{d\bar{x}}{dt} = \bar{f}(\bar{x}, t), \qquad \text{where } \bar{f} : E \times \mathbb{R} \to E \text{ is a vector field.}$$

If $\dfrac{\partial \bar{f}}{\partial t} = 0$ the system is autonomous.

A non-autonomous system reduces to an autonomous by defining $\bar{y} = (\bar{x}, t) \in E \times \mathbb{R}$ so that

$$\dot{\bar{y}} = \begin{pmatrix} \bar{f}(\bar{y}) \\ 1 \end{pmatrix}.$$

Thus, can study only autonomous systems.

A nth order ODE

$$\frac{d^n x}{dt^n} = g\left(x, \frac{dx}{dt}, \ldots, \frac{d^{n-1}x}{dt^{n-1}}\right)$$

can be reduced to a standard form with n equations by introducing

$$\bar{y} = \left(x, \frac{dx}{dt}, \ldots, \frac{d^{n-1}x}{dt^{n-1}}\right) = (y_1, y_2, \ldots, y_n)$$

so that

$$\dot{\bar{y}} = \begin{pmatrix} y_2 \\ y_3 \\ \vdots \\ y_n \\ g(y_1, y_2, \ldots, y_n) \end{pmatrix}.$$

Similarly to autonomous systems, will only consider maps of the form

$$\bar{x}_{n+1} = \bar{F}(\bar{x}_n) \qquad \text{and not} \qquad \bar{F}(\bar{x}_n, n).$$

1.2 Initial value problem (IVP)

We would like to explore an initial value problem for $\dot{\bar{x}} = \bar{f}(\bar{x}, t)$:

> Given the initial value of $\bar{x}$: $\bar{x}(t_0) = \bar{x}_0$, $\bar{x}_0 \in E$, $t \in I \subseteq \mathbb{R}$, find a differentiable function $\bar{x}(t)$ for $t \in I$ which remains in E for $t \in I$ and satisfies the initial condition and the ODE.

For an autonomous system can define solution in terms of a flow ϕ_t.

Definition (Flow) $\phi_t(\bar{x})$ such that $\phi_t(\bar{x}_0)$ is the solution of $\dot{\bar{x}} = \bar{f}(\bar{x})$ starting at $\bar{x}_0$ when $t = 0$ is called the flow through $\bar{x}_0$ at $t = 0$.

Thus,

$$\phi_0(\bar{x}_0) = \bar{x}_0, \qquad \phi_s\big(\phi_t(\bar{x}_0)\big) = \phi_{s+t}(\bar{x}_0).$$

Does this solution exist? Is it unique?

Theorem (Cauchy–Peano)

If $\bar{f}(\bar{x}, t)$ is continuous and $|\bar{f}| < M$ in the domain

$$D : \big\{|t - t_0| < \alpha,\ |\bar{x} - \bar{x}_0| < \beta\big\}$$

then the IVP has a solution for $|t - t_0| < \min(\alpha, \beta/M)$.

But uniqueness requires stronger conditions on f.

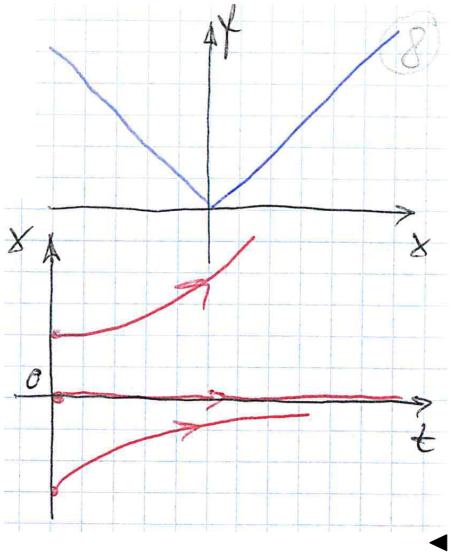
Example Unique solution

$$\dot{x} = |x|, \qquad x(t_0) = x_0$$

Solution

$$x(t) = \begin{cases} x_0 e^{t_0 - t}, & x_0 < 0, \\ 0, & x_0 = 0, \\ x_0 e^{t - t_0}, & x_0 > 0. \end{cases}$$

Note that f is continuous but not differentiable.

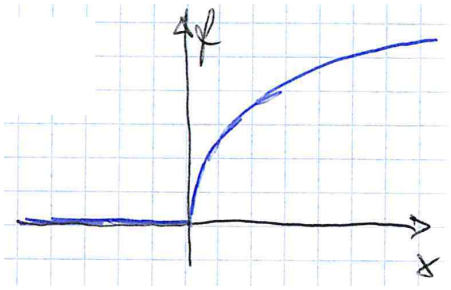


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Example Non-unique solution

$$\dot{x} = \begin{cases} \sqrt{x}, & x \geq 0, \\ 0, & x < 0, \end{cases} \qquad x(0) = 0$$

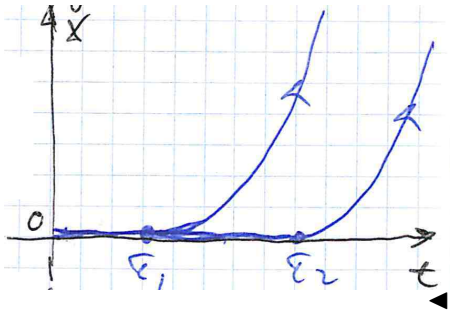
Can check that



$$x(t) = \begin{cases} 0, & t < \tau, \\ \frac{1}{4}(t - \tau)^2, & t \geq \tau \end{cases} \qquad \text{for any } \tau \geq 0$$

satisfies the equation and initial condition. But τ can be anything so a family of solutions!

Here again f is continuous and not differentiable.



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Why is the difference? Because $\dfrac{d}{dx}(\sqrt{x})$ is not bounded at 0!

To guarantee uniqueness, need f to be not just continuous, but also Lipschitz.

Definition Lipschitz property

A function $\bar{f}$ defined on a subset of $\mathbb{R}^n$ satisfies Lipschitz condition at a point $\bar{x}_0$ with Lipschitz constant L if $\exists\,(L, a)$ such that $\forall\,\bar{x}, \bar{y}$: $|\bar{x} - \bar{x}_0| < a$, $|\bar{y} - \bar{x}_0| < a$ we have

$$|\bar{f}(\bar{x}) - \bar{f}(\bar{y})| \leq L|\bar{x} - \bar{y}|.$$

Note that Differentiable $\rightarrow$ Lipschitz $\rightarrow$ Continuous

Theorem Uniqueness/Picard–Lindelöf/Cauchy–Lipschitz

Consider an IVP $\dot{\bar{x}} = \bar{f}(\bar{x}, t)$ with $\bar{x} = \bar{x}_0$ at $t = t_0$. If $\bar{f}$ satisfies a Lipschitz condition at $\bar{x}_0$ then the solution $\phi_{t-t_0}(\bar{x}_0)$ exists and is unique and continuous in a neighborhood of $(\bar{x}_0, t_0)$.

However, in general uniqueness & continuity do not necessarily mean existence of solution for all time!

Example Finite time blowup

$$\dot{x} = x^2, \qquad x(0) = 1, \qquad \text{solution is } x(t) = \frac{1}{1-t},$$

blows up to ∞ at $t \rightarrow 1$.

In what follows, unless stated otherwise, we will consider only differentiable functions f.

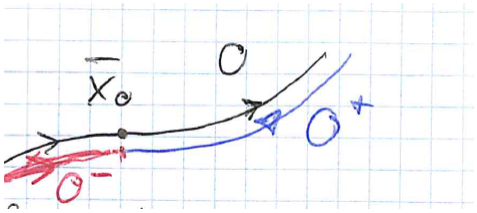
§1.3. Orbits, invariant sets and limit sets

Consider a flow $\phi_t(\bar{x}_0)$.

Definition

* The orbit/trajectory of ϕ_t through $\bar{x}_0$ is the set

$$O(\bar{x}_0) = \{\phi_t(\bar{x}_0) : -\infty < t < \infty\}.$$



* The forward orbit of ϕ_t through $\bar{x}_0$ is the set

$$O^+(\bar{x}_0) = \{\phi_t(\bar{x}_0) : 0 \le t\};$$

analogously, backward orbit $O^-(\bar{x}_0)$ if for $t \le 0$.

Definition

A set of points $\mathcal{M} \subseteq E$ is invariant under f if

$$\bar{x} \in \mathcal{M} \quad \Rightarrow \quad O(\bar{x}) \subseteq \mathcal{M}.$$

Clearly, any orbit is invariant.

Definition Fixed point

The point $\bar{x}_0$ is a fixed point (equilibrium, stationary, critical point) if $f(\bar{x}_0) = 0$. Then $\bar{x} = \bar{x}_0$ for any t and $O(\bar{x}_0) = \bar{x}_0$.

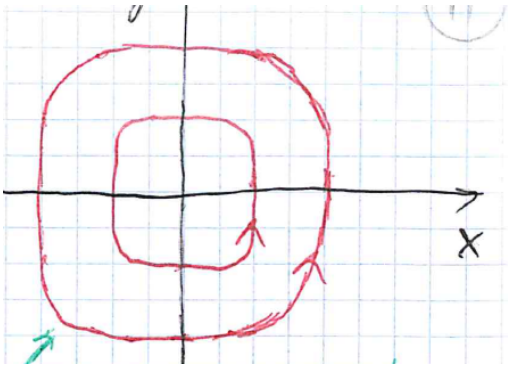
Definition Periodic point & orbit

A point $\bar{x}_0$ is a periodic point with period T if $\phi_T(\bar{x}_0) = \bar{x}_0$ for some $T > 0$ but $\phi_t(\bar{x}_0) \neq \bar{x}_0$ for $0 < t < T$.

The set $\{\phi_t(\bar{x}_0) : 0 \le t < T\}$ is the periodic orbit through $\bar{x}_0$.

Example Periodic orbit

$$\begin{pmatrix}\dot{x}\\ \dot{y}\end{pmatrix} = \begin{pmatrix}-y^3\\ x^3\end{pmatrix} \quad \Rightarrow \quad \underline{x^4 + y^4 = \text{cst}(t)}$$



not a limit cycle!

Definition Limit cycle

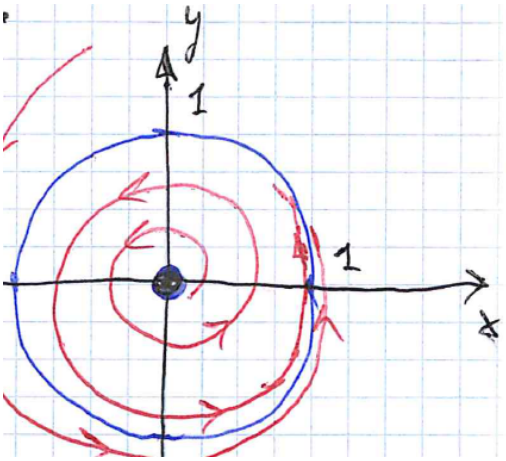
A limit cycle is an isolated periodic orbit, i.e. there are no other periodic orbits in its sufficiently small neighborhood.

Example Limit cycle

$$\begin{pmatrix}\dot{x}\\ \dot{y}\end{pmatrix} = \begin{pmatrix}-y + x(1-x^2-y^2)\\ x + y(1-x^2-y^2)\end{pmatrix} \quad \Rightarrow$$

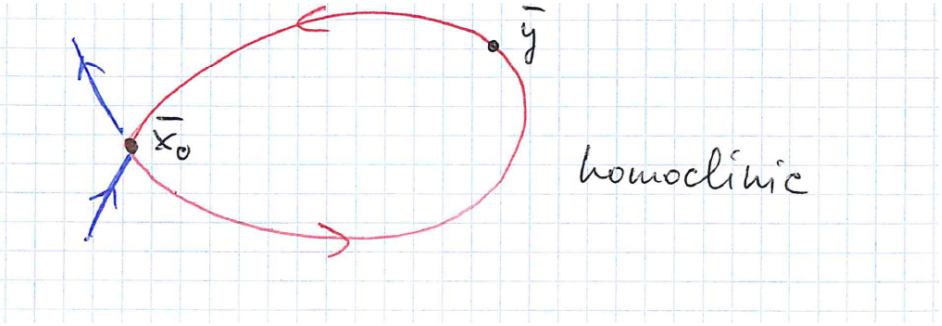
$$\underline{\dot{r} = r(1-r^2)}, \qquad \underline{\dot{\theta} = 1}.$$

A fixed point $(0,0)$ and a limit cycle $r = 1$.



Definition Homoclinic orbit

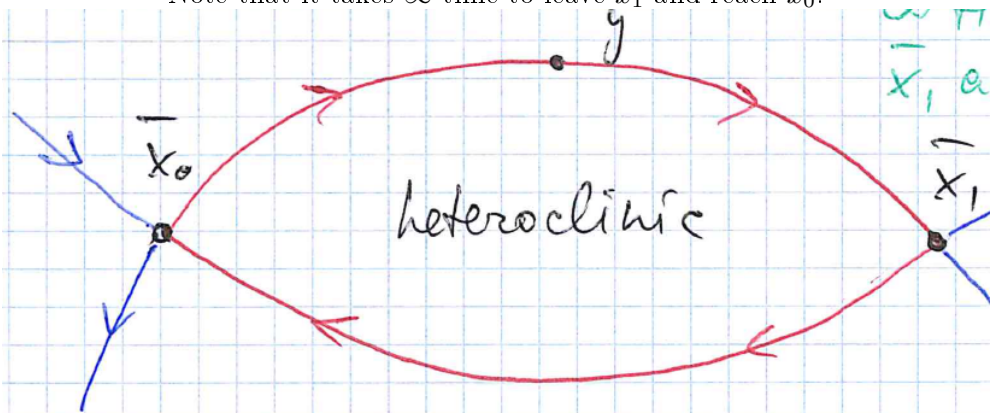
If $\bar{x}_0$ is a fixed point and $\exists\, \bar{y} \neq \bar{x}_0$ such that $\phi_t(\bar{y}) \to \bar{x}_0$ as $t \to \infty$ and $t \to -\infty$, then $O(\bar{y})$ is a homoclinic orbit.



Definition Heteroclinic orbit

If $\bar{x}_0$ and $\bar{x}_1 \neq \bar{x}_0$ are fixed points and $\exists\, \bar{y}$ such that $\phi_t(\bar{y}) \to \bar{x}_0$ as $t \to \infty$ and $\phi_t(\bar{y}) \to \bar{x}_1$ at $t \to -\infty$, then $O(\bar{y})$ is a heteroclinic orbit.

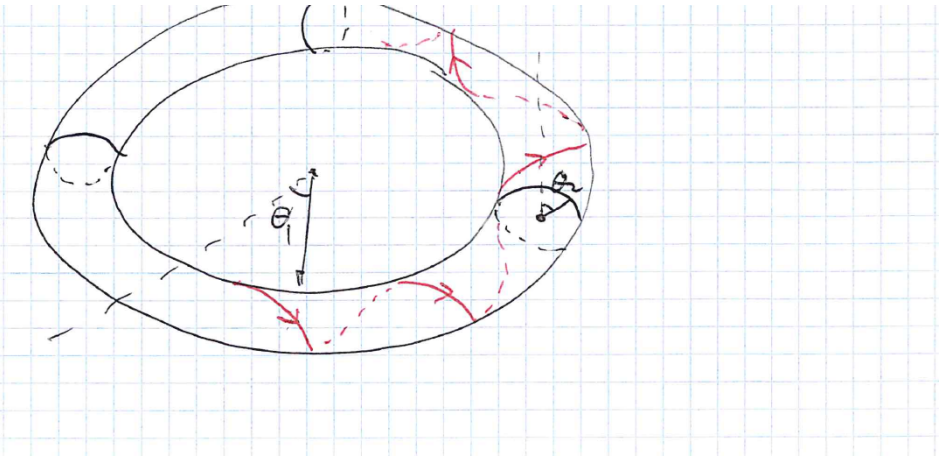
Note that it takes ∞ time to leave $\bar{x}_1$ and reach $\bar{x}_0$!



In higher dimensions more complex invariant sets are possible.

Example 2-torus in 3D

θ_1, θ_2 coordinates on the surface of a torus in 3D, $\dot{\theta}_1 = \omega_1$, $\dot{\theta}_2 = \omega_2$ and ω_1, ω_2 are not in a ratio, $\Rightarrow$ trajectory will fill the full surface of a torus. Thus, surface is an invariant set.



When talking about the long-term behavior of trajectories need to be careful with statements like $\phi_t(\bar{x}) \to \bar{y}$ as such limit may not exist, e.g. for periodic orbits.

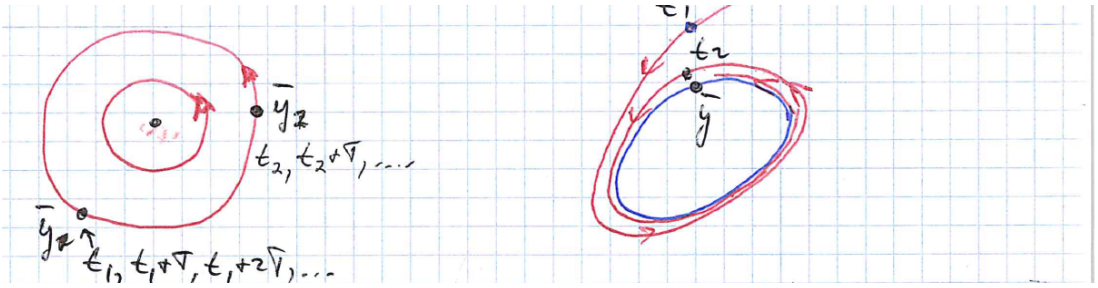
Instead use

Definition

ω-limit set of $\bar{x}$ is

$$\omega(\bar{x}) = \{\bar{y} : \exists \text{ some sequence } t_1, t_2, \ldots \to \infty \text{ with } \phi_{t_n}(\bar{x}) \to \bar{y}\}\,.$$

α-limit set defined similarly but for $t \to -\infty$.



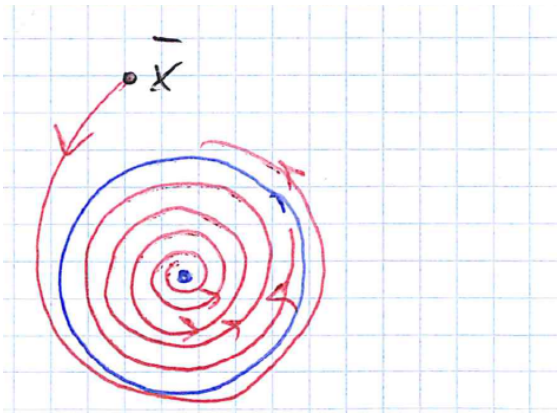
When the orbit $O(\bar{x})$ is bounded, $\omega(\bar{x})$ has properties:

* non-empty
* closed
* connected
* invariant under f
* bounded

Example

Go back to

$$\dot{r} = r(1 - r^2), \qquad \dot{\theta} = 1.$$



Here $\omega(\bar{x}) = \{r = 1\}$,

$\alpha(\bar{x}) = \{(0,0)\}$ for $|\bar{x}| < 1$ or undefined for $|\bar{x}| > 1$.

§1.4. Topological equivalence of flows

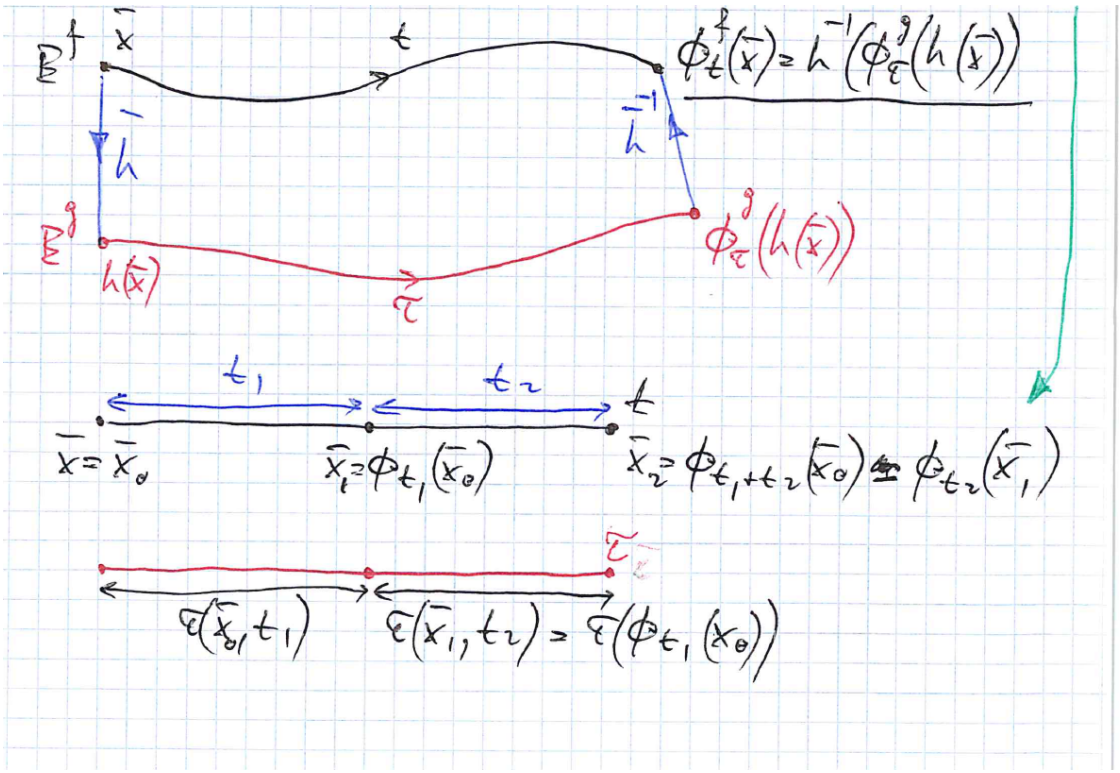
In this course we will sometimes be referring to different dynamical systems/flows as topologically/geometrically equivalent. Later we'll see that the flow structure changes at bifurcation.

Definition Topological equivalence

Two flows $\phi_t^f(\bar{x})$ and $\phi_t^g(\bar{x})$ are topologically equivalent if there is a homeomorphism $h(\bar{x}) : E^f \to E^g$ (i.e. a continuous bijection with continuous inverse) and time-increasing function $\tau(\bar{x}, t)$ such that for any $t, \bar{x}$

$$\phi_t^f(\bar{x}) = h^{-1} \circ \phi_\tau^g \circ h(\bar{x})$$

$$\text{and} \qquad \tau(\bar{x}, t_1 + t_2) = \tau(\bar{x}, t_1) + \tau\big(\phi_{t_1}^f(\bar{x}), t_2\big).$$



End of lecture 3

§1.4. Topological equivalence of flows

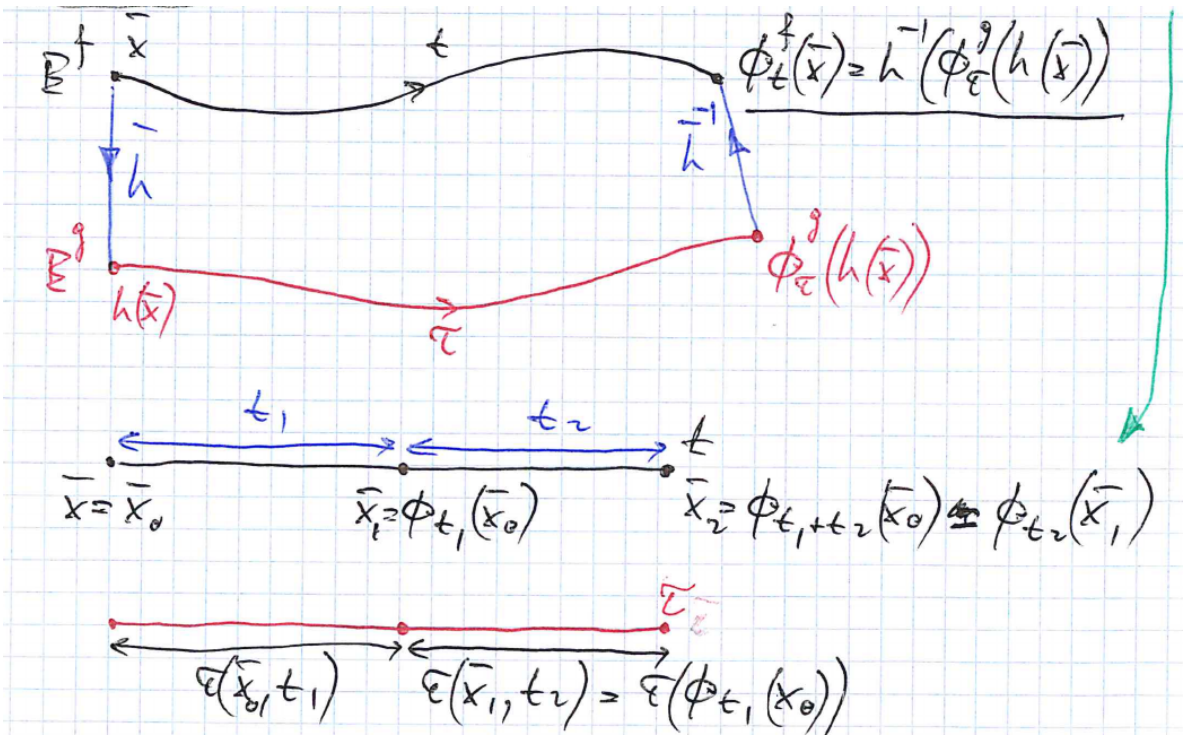
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$$\phi_t^f(\bar{x}) = \bar{h}^{-1} \circ \phi_\tau^g \circ h(\bar{x})$$

and $$\tau(\bar{x}, t_1 + t_2) = \tau(\bar{x}, t_1) + \tau\big(\phi_{t_1}^f(\bar{x}), t\big).$$



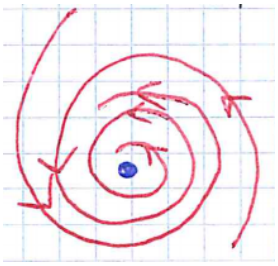
In words, $\exists$ a 'nice' mapping from one phase-space to the other and a map from one time to another such that the evolutions of the two systems are the same.

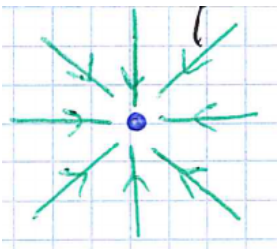
Clearly, $\bar{h}$ must map fixed points into fixed points and periodic orbits to periodic orbits (with generally different period).

If $\tau = t$ we say that the two flows are topologically conjugate.

Example

$$\begin{aligned} \dot{r} &= -r \\ \dot{\theta} &= 1 \end{aligned} \qquad \text{and} \qquad \begin{aligned} \dot{\rho} &= -2\rho \\ \dot{\psi} &= 0 \end{aligned}$$





Flows are topologically equivalent with $\bar{h}(0,0) = (0,0)$, $h(r,\theta) = (r^2, \theta + \ln r)$ for $r \neq 0$ in polars and $\tau = t$ (so topologically conjugate).

To show, solve both: starting at $\bar{x} = (r_0, \theta_0)$

$$\phi_t^f = (r_0 e^{-t}, \theta_0 + t), \qquad \phi_t^g = (\rho_0 e^{-2t}, \psi_0).$$

Then

$$h \circ \phi_t^f = \left(r_0^2 e^{-2t}, \theta_0 + t + \ln(r_0 e^{-t})\right) = \left(r_0^2 e^{-2t}, \theta_0 + \ln r_0\right)$$

and

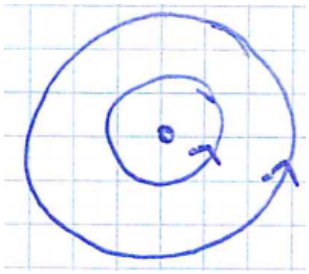
$$\phi_t^g \circ h = \left(r_0^2 e^{-2t}, \theta_0 + \ln r_0\right) \quad \text{– the same, } \Rightarrow \text{ Ok}$$

$$\text{as } h(\bar{x}) = (r_0^2, \theta_0 + \ln r_0).$$

◀

Example

$$\begin{aligned}\dot{r} &= 0 \\ \dot{\theta} &= 1\end{aligned} \qquad \text{and} \qquad \begin{aligned}\dot{r} &= 0 \\ \dot{\theta} &= r + \sin^2\theta\end{aligned}$$



Both are periodic circular orbits, so $\bar{h}(\bar{x}) = \bar{x}$, and time is stretched:

$$\frac{d\theta}{dt} * (r + \sin^2\theta) = \frac{d\theta}{d\tau}, \quad \Rightarrow$$

$$\tau = \int_0^t \frac{dt'}{r_0 + \sin^2(\theta_0 + t')}$$

as first flow is

$$\phi_t(r_0, \theta_0) = (r_0, \theta_0 + t).$$

Thus, topologically equivalent (but not conjugate). ◀

Definition

The vector field $\bar{f}$ (or system $\dot{\bar{x}} = \bar{f}(\bar{x})$ or flow $\phi_t(\bar{x})$) is structurally stable if $\exists\, \varepsilon > 0$ such that $\bar{f} + \bar{\delta}$ is topologically equivalent to $f\ \forall\, \bar{\delta}(\bar{x})$ with

$$|\bar{\delta}| + \sum_i \left| \frac{\partial \bar{\delta}}{\partial x_i} \right| < \varepsilon.$$

In the first example system is structurally stable, in the second it is not (a small $\delta \neq 0$ will turn $\dot{r} = 0$ into $\dot{r} = \delta$, which is no longer a periodic orbit).

We will discuss this later in the context of bifurcations.

§2. Behavior near fixed points

Suppose $\bar{x}_0$ is a fixed point of $\dot{\bar{x}} = \bar{f}(\bar{x})$, i.e. $\bar{f}(\bar{x}_0) = 0$. If f is smooth near $\bar{x}_0$, set

$$\bar{y} = \bar{x} - \bar{x}_0, \quad \Rightarrow \quad \dot{\bar{y}} = (\bar{x} - \dot{\bar{x}}_0) = \dot{\bar{x}} = \bar{f}(\bar{x}) = \bar{f}(\bar{y} + \bar{x}_0).$$

Taylor expand r.h.s.:

$$\dot{\bar{y}} = \underbrace{\bar{f}(\bar{x}_0)}_{=0} + \hat{A}\bar{y} + O(|y|^2), \qquad A_{ij} = \frac{\partial f_i}{\partial x_j} \quad \begin{matrix} \text{– the Jacobian} \\ \text{matrix of } f \text{ at } \bar{x}_0. \end{matrix}$$

Thus, near $\bar{x}_0$ the behavior of the system is approximated by $\dot{\bar{y}} = \hat{A}\bar{y}$.

§2.1 2D flows near fixed points

In 2D is a 2×2 matrix $\hat{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with $\mathrm{Det}(\hat{A}) = ad - bc$, $\mathrm{Tr}(\hat{A}) = a + d$.

Eigenvalues satisfy characteristic eq-n

$$\lambda^2 - \lambda \cdot \mathrm{Tr}(\hat{A}) + \det(\hat{A}) = 0, \quad \Rightarrow \quad \lambda = \frac{1}{2}\left[\mathrm{Tr}(\hat{A}) \pm \sqrt{(\mathrm{Tr}(\hat{A}))^2 - 4\,\mathrm{Det}(\hat{A})}\right].$$

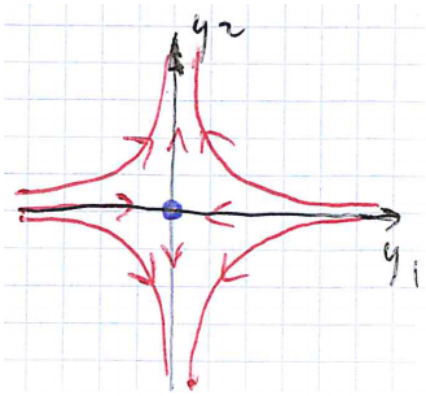
By finding eigenvectors (when we can) we can put $\hat{A}$ into canonical form & analyze.

① Saddle point: $\mathrm{Det}(\hat{A}) < 0$, $\Rightarrow \lambda_1, \lambda_2$ are real and have opposite signs, $\lambda_1\lambda_2 < 0$.

$$\hat{A} = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}, \quad \text{e.g. } \lambda_1 < 0 < \lambda_2$$

$$\Updownarrow$$

$$\dot{y}_1 = \lambda_1 y_1, \quad \dot{y}_2 = \lambda_2 y_2, \quad \Rightarrow \quad \begin{matrix} y_1 = y_{1,0} e^{\lambda_1 t} \\ y_2 = y_{2,0} e^{\lambda_2 t} \end{matrix}$$



End of lecture 4

Since, eliminating t, $y_2 \propto y_1^{\lambda_2/\lambda_1} = \dfrac{1}{y_1^{|\lambda_2/\lambda_1|}}$. So, orbits are hyperbolas!

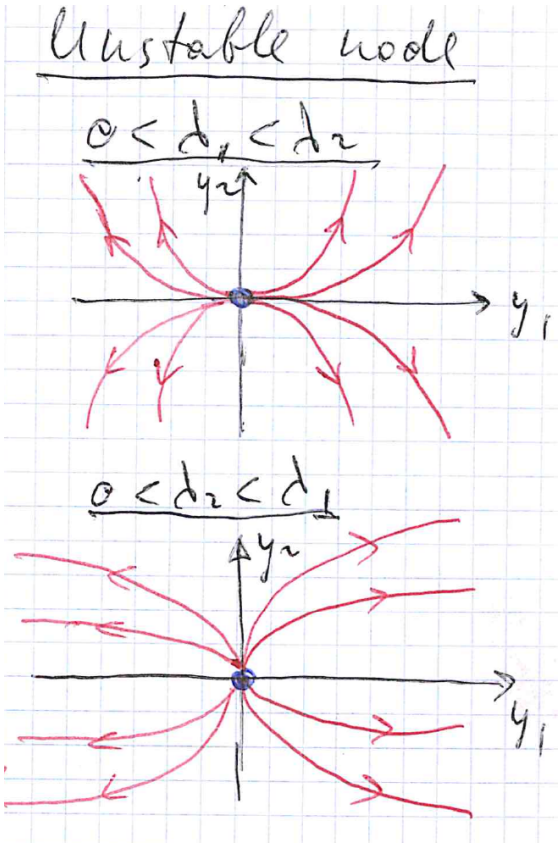
② Node: $(\mathrm{Tr}(\hat{A}))^2 > 4\,\mathrm{Det}(\hat{A}) > 0$, $\Rightarrow \lambda_1, \lambda_2$ are real and either both positive ($\mathrm{Tr}(\hat{A}) > 0$) or negative ($\mathrm{Tr}(\hat{A}) < 0$).

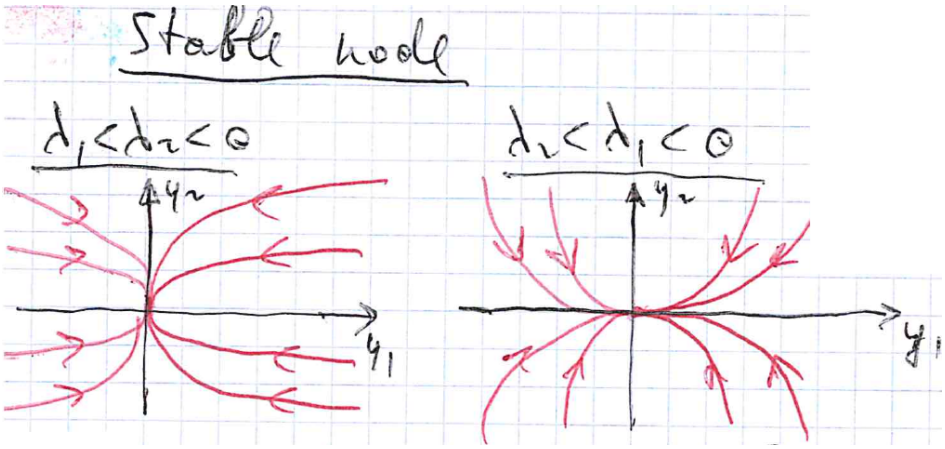
Same canonical form

$$\hat{A} = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}, \quad \Rightarrow \quad y_2 \propto y_1^{\lambda_2/\lambda_1}.$$

Get different patterns depending on $\lambda_2/\lambda_1 < 1$ or > 1.

Sign of λ_1, λ_2 sets the direction of the flow.





③ Focus (spiral): $(\mathrm{Tr}(\hat{A}))^2 < 4\,\mathrm{Det}(\hat{A})$ and $\det(\hat{A}) > 0$ so that roots are complex, $\lambda = \rho \pm i\omega$, $\omega \neq 0$.

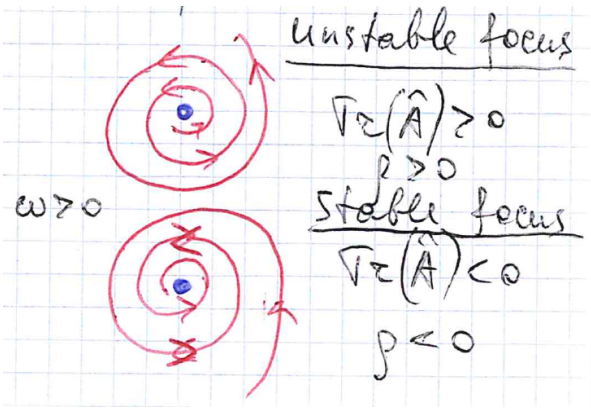
Canonical form

$$\hat{A} = \begin{pmatrix} \rho & -\omega \\ \omega & \rho \end{pmatrix}.$$

In polars have

$$\dot{r} = \rho r, \quad \dot{\theta} = \omega, \quad \Rightarrow$$

$$r = r_0 e^{\rho t}, \quad \theta = \theta_0 + \omega t.$$



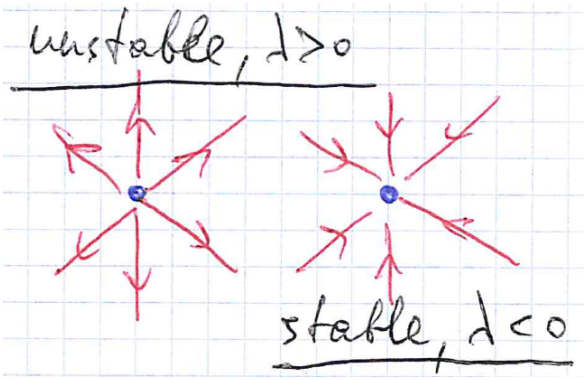
Other cases are degenerate.

④ $(\mathrm{Tr}(\hat{A}))^2 = 4\,\mathrm{Det}(\hat{A}) > 0$

A boundary between focus & node, double eigenvalue $\lambda = \frac{1}{2}\mathrm{Tr}(\hat{A})$. Two cases:

Stellar node (star)

if $\hat{A} = \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}$ is canonical



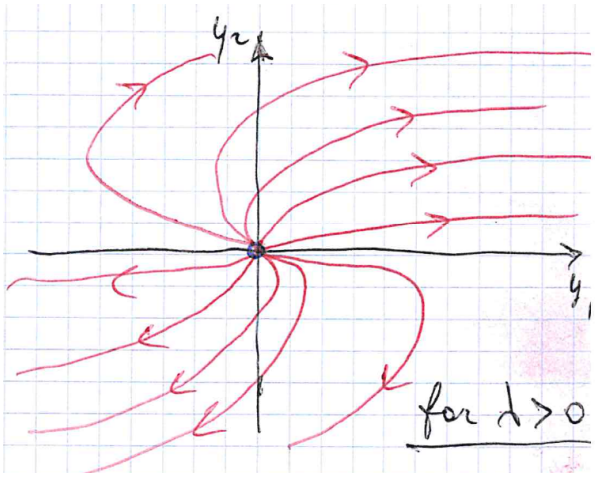
Improper node

if $\hat{A} = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix}$ is canonical

(defective)

Here $y = y_0 e^{\lambda t}$,

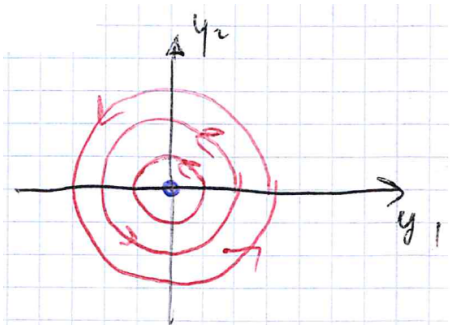
$x = x_0 e^{\lambda t} + y_0 t e^{\lambda t}$, $\quad \dfrac{y}{x} = \dfrac{y_0}{x_0 + y_0 t}$.



⑤ $\mathrm{Det}(\hat{A}) > 0$, $\quad \mathrm{Tr}(\hat{A}) = 0$: Centre

Here $\lambda = \pm i\omega$, $\qquad \hat{A} = \begin{pmatrix} 0 & -\omega \\ \omega & 0 \end{pmatrix}$.

$\dot{r} = 0$, $\quad \dot{\theta} = \omega$, $\quad \Rightarrow$ orbits are closed circles.

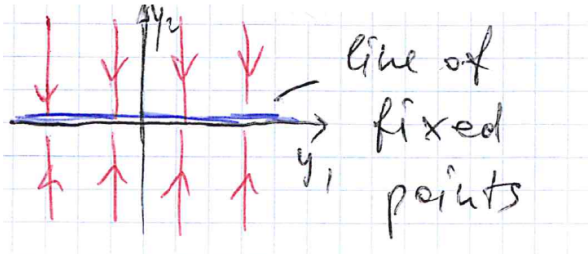


⑥ $\mathrm{Det}(\hat{A}) = 0$, $\quad \mathrm{Tr}(\hat{A}) \neq 0$ – between saddle & node

$\lambda_1 = 0$, $\quad \lambda_2 \neq 0$, $\qquad \hat{A} = \begin{pmatrix} 0 & 0 \\ 0 & \lambda \end{pmatrix}$

$y_1 = \text{const}$, $\qquad y_2 = y_{2,0} e^{\lambda t}$

for $\lambda < 0$

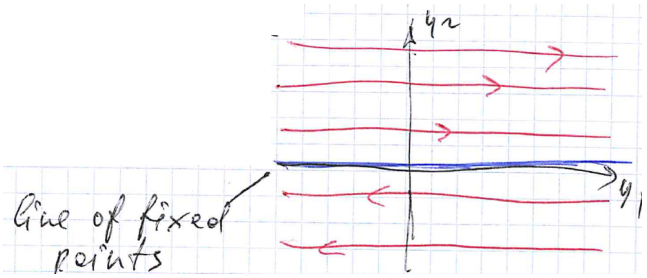


⑦ $\underline{\mathrm{Det}(\hat{A}) = 0, \quad \mathrm{Tr}(\hat{A}) = 0}$, $\lambda_{1,2} = 0$

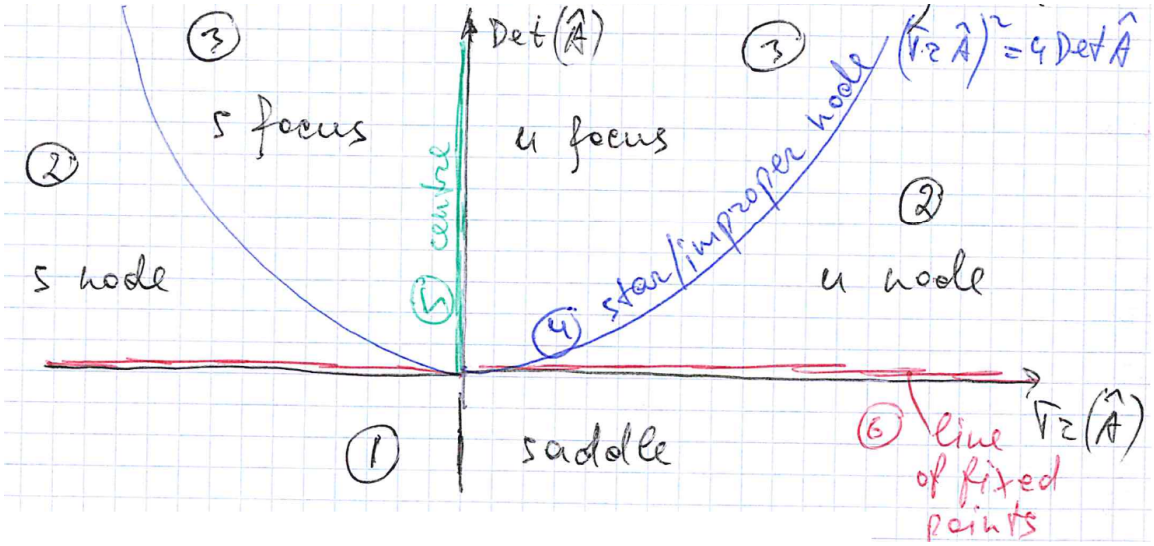
if $\hat{A} = 0$, $\Rightarrow y_1 = \mathrm{const}$, $y_2 = \mathrm{const}$, $\Rightarrow$ plane of fixed points!

if $\hat{A} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, this is a case of shear,

$$\underline{\dot{y}_1 = y_2, \quad \dot{y}_2 = 0.}$$



This zoo can be summarized in a diagram



Canonical forms are in special basis

$$(*) \quad \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} \quad \text{– eigenvectors}$$

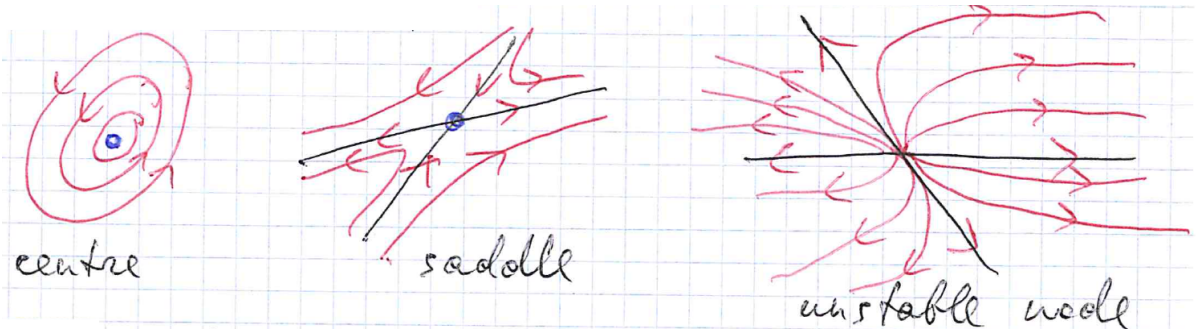
$$(*) \quad \begin{pmatrix} \lambda & 1 \\ 0 & \lambda \end{pmatrix} \quad \text{– generalized eigenvectors}$$

$$(*) \quad \begin{pmatrix} \rho & -\omega \\ \omega & \rho \end{pmatrix} \quad \text{– Re \& Im of eigenvectors}$$

Notes

(∗) No need to change basis in practice, just work with $\mathrm{Tr}(\hat{A})$, $\mathrm{Det}(\hat{A})$ to classify

(∗) In original (non-canonical) coordinates shapes can be distorted



(∗) Useful to know eigenvectors for saddles/nodes; for focus/centre cases ratios depend also on $\lambda_{1,2}$, easier to just plot several values

(∗) Centres are rare but generic in Hamiltonian systems:

$$\dot{\bar{x}} = \begin{pmatrix} \dfrac{\partial H}{\partial y} \\ -\dfrac{\partial H}{\partial x} \end{pmatrix} \quad \text{for some } H(x,y), \quad \bar{x} = (x,y).$$

$$\text{Indeed,} \quad \hat{A} = \begin{pmatrix} H_{xy} & H_{yy} \\ -H_{xx} & -H_{xy} \end{pmatrix}, \quad \Rightarrow \quad \underline{\mathrm{Tr}(\hat{A}) = 0}, \quad \Rightarrow$$

$$\text{either} \quad \underset{\lambda = \pm i\omega}{\text{centres}} \quad \text{or} \quad \underset{\lambda_1 \lambda_2 < 0}{\text{saddles}}.$$

$$\text{Also,} \quad \underline{\frac{dH}{dt}} = (\dot{\bar{x}} \cdot \nabla) H = \dot{x} H_x + \dot{y} H_y = H_y H_x - H_x H_y = 0.$$

Thus, $H(x,y) = \text{const}$ on trajectories, orbits follow contours of $H(x,y)$ in autonomous Hamiltonian systems.

End of lecture 5

§2.2 Effect of nonlinear terms

How does the behavior of linearization $\dot{\bar{y}} = \hat{A}\bar{y}$ of the system $\dot{\bar{x}} = \bar{f}(\bar{x})$ about the fixed point $\bar{x}_0$ inform us about the behavior of the full nonlinear system around $\bar{x}_0$?

Definition

A fixed point $\bar{x}_0$ is hyperbolic if none of the eigenvalues of the Jacobian of $\bar{f}(\bar{x})$ (matrix $\hat{A}$) at $\bar{x}_0$ have zero real part. It is non-hyperbolic otherwise.

We classify hyperbolic fixed points as

- $(*)$ source if all $\mathrm{Re}(\lambda_i) > 0$
- $(*)$ sink if all $\mathrm{Re}(\lambda_i) < 0$
- $(*)$ saddle otherwise (i.e. some $\mathrm{Re}(\lambda_i) > 0$, others < 0).

We will discuss separately hyperbolic & non-hyperbolic fixed points.

One can show (e.g. see Glendinning, Ch. 4) that if

(1) $\bar{x}_0$ is hyperbolic and

(2) the nonlinear corrections of $\bar{f}$ are $O(|\bar{x} - \bar{x}_0|^2)$

then the full nonlinear system and the linearized system are topologically conjugate.

Note: non-hyperbolic fixed points arise when

$$\{\mathrm{Det}\,\hat{A} \geq 0,\ \mathrm{Tr}\,\hat{A} = 0\}, \quad \{\mathrm{Det}\,\hat{A} = 0,\ \mathrm{Tr}\,\hat{A} \neq 0\},$$

i.e. cases 5, 6, 7 identified earlier.

§2.2.1 Stable and unstable manifolds

For a linearized system we can separate the phase space into different domains according to different behaviors in time.

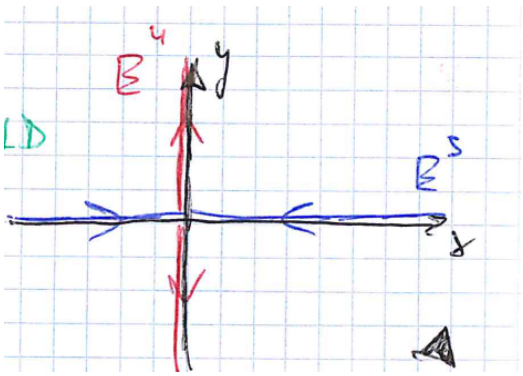
Definition

The stable, unstable and centre subspaces of the linearization of $\bar{f}(\bar{x})$ near a fixed point $\bar{x}_0$ are the three linear subspaces E^s, E^u and E^c spanned by subsets of (possibly generalized) eigenvectors of $\hat{A}$ whose eigenvalues have real parts $\mathrm{Re}(\lambda) < 0$, $\mathrm{Re}(\lambda) > 0$ and $\mathrm{Re}(\lambda) = 0$, respectively.

Note: a hyperbolic fixed point has no centre subspace.

Example

$$\underset{\text{2D}}{\hat{A} = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}}, \quad \Rightarrow \quad \begin{matrix} E^s = x\text{-axis} \\ E^u = y\text{-axis} \end{matrix} \quad \text{1D}$$



Theorem Stable manifold theorem

Suppose $\bar{0}$ is a hyperbolic fixed point of $\dot{\bar{x}} = \bar{f}(\bar{x})$ and E^s and E^u are the stable and unstable subspaces of the linearization of $\bar{f}$ near $\bar{0}$. Then $\exists$ stable and unstable manifolds W^s_{loc} & W^u_{loc}, which have the same dimension as E^s, E^u and are tangent to E^s & E^u at $\bar{0}$, such that for $\bar{x} \neq 0$ but in a sufficiently small neighborhood of $\bar{0}$,

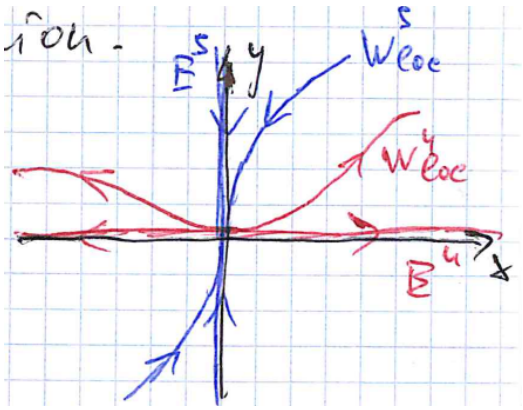
$$W^s_{\text{loc}} = \{\bar{x} : \phi_t(\bar{x}) \to \bar{0} \text{ as } t \to +\infty\},$$

$$W^u_{\text{loc}} = \{\bar{x} : \phi_t(\bar{x}) \to \bar{0} \text{ as } t \to -\infty\}.$$

Proof: not part of the course, see Glendinning.

The local stable manifold W^s_{loc} can be extended to a global invariant manifold W^s by following the flow backwards in time starting at $\bar{x} \in W^s_{\text{loc}}$. Same for W^u_{loc}, extend to W^u by $t \to +\infty$.

In $\mathbb{R}^2$, this theorem guarantees the existence of two trajectories that approach and leave a saddle fixed point in a tangent fashion.



It is often easy in practice to find local approximations to stable and unstable manifolds.

Indeed, for a saddle in $\mathbb{R}^2$ assume that E^s is $x = 0$ and E^u is $y = 0$ (can always change the basis).

Then look for W^s_{loc} in the form $x = S(y)$ with $S(0) = 0$ and $S'(0) = 0$. For W^u_{loc} take $y = U(x)$ with $U(0) = 0$, $U'(0) = 0$. These manifolds must satisfy $\dot{\bar{x}} = \bar{f}(\bar{x})$, i.e. $\dot{x} = \frac{\partial S}{\partial y}\dot{y}$ and $\dot{y} = \frac{\partial U}{\partial x}\dot{x}$ must solve $\dot{\bar{x}} = \bar{f}(\bar{x})$. Thus, for

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \begin{pmatrix} f(x,y) \\ g(x,y) \end{pmatrix},$$

$$\dot{x} = \frac{dS}{dy}\dot{y} = \underbrace{\frac{dS}{dy} g(S(y),y) = f(S(y),y)}_{\text{(nonlinear) ODE for } S(y)} \quad \text{and}$$

$$\dot{y} = \frac{dU}{dx}\dot{x} = \underbrace{\frac{dU}{dx} f(x,U(x)) = g(x,U(x))}_{\text{ODE for } U(x)}.$$

No need to solve these ODEs, we can simply find a locally convergent series expansion in the form

$$S(y) = a_2 y^2 + a_3 y^3 + \cdots, \qquad U(x) = b_2 x^2 + b_3 x^3 + \cdots$$

and solve term by term.

Example

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \begin{pmatrix} x(1-y) \\ -y + x^2 \end{pmatrix}, \qquad \hat{A} = \begin{pmatrix} 1-y & -x \\ 2x & -1 \end{pmatrix}\Big|_{(0,0)} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

$$\det \hat{A} = -1 \;\Rightarrow\; \underline{\text{saddle}}, \qquad \Rightarrow \quad E^s = \{x = 0\}, \quad E^u = \{y = 0\}.$$

For W^u_{loc} take

$$y = U(x) = a_2 x^2 + a_3 x^3 + a_4 x^4 + \cdots,$$

substitute into

$$\dot{y} = -y + x^2 = \frac{dU}{dx} x(1-y), \quad \Rightarrow$$

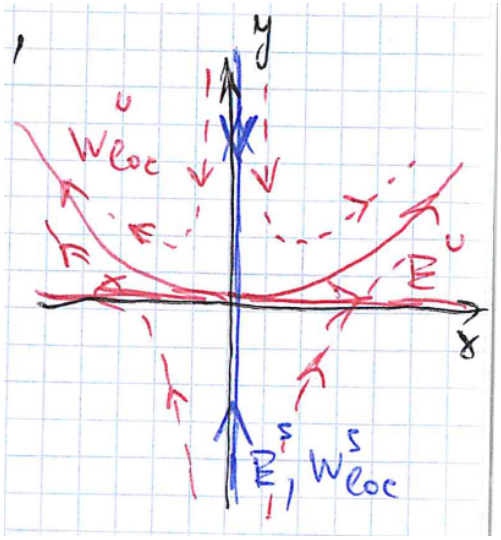
$$-U(x) + x^2 = \frac{dU}{dx} x(1 - U(x)), \quad \Rightarrow$$

$$x^2 - a_2 x^2 - a_3 x^3 - a_4 x^4 + \cdots = (2a_2 x + 3a_3 x^2 + 4a_4 x^3 + \cdots)\, x \times (1 - a_2 x^2 - a_3 x^3 - a_4 x^4 + \cdots), \quad \Rightarrow$$

$$\begin{aligned} x^2: &\quad 1 - a_2 = 2a_2 &&\Rightarrow \quad a_2 = \frac{1}{3} \\ x^3: &\quad -a_3 = 3a_3 &&\Rightarrow \quad a_3 = 0 \\ x^4: &\quad -a_4 = 4a_4 - 2a_2^2 &&\Rightarrow \quad a_4 = \frac{2}{45} \end{aligned}$$

due to $x \mapsto -x$ symmetry! $\quad (a_3 = 0)$

$$\text{Thus,} \quad U(x) = \frac{x^2}{3} + \frac{2}{45} x^4 + \cdots.$$



To find W^s_{loc}, $x = S(y) = a_2 y^2 + a_3 y^3 + \cdots$,

$$S(y)(1-y) = \frac{dS}{dy}[-y + S^2(y)],$$

repeat calculation, find $a_n = 0\ \forall n$. This is caused by symmetry! If $X = -x$ then system the same so W^s_{loc} has to be symmetric w.r.t. y-axis & tangent to it,

$$\Rightarrow \quad W^s_{\text{loc}} = y\text{-axis}.$$

End of lecture 6

Note:

- If E^s & E^u are not aligned with the coordinate axes, one can either transform the coordinates to get alignment, or consider functions S & U tangent to the relevant eigenvector. E.g. for $\bar{e} = \binom{1}{k}$ use $y = kx + a_2x^2 + a_3x^3 + \cdots$.
- In higher dimensions use multi-coordinate versions of S & U. E.g. for a 2D E^s in 3D use S: $z = S(x, y)$ and $\dot{z} = \dot{x}\dfrac{\partial S}{\partial x} + \dot{y}\dfrac{\partial S}{\partial y}$ and expand in both coordinates x, y.

§2.2.2 Non-linear terms in non-hyperbolic cases

Suppose at least one eigenvalue has $\mathrm{Re}(\lambda) = 0$. Have many possibilities.

(a) $\lambda = \pm i\omega$, linear system has a centre. In nonlinear case multiple possible outcomes, e.g.

* $\begin{aligned}\dot{x} &= -y - x^3,\\ \dot{y} &= x - y^3,\end{aligned}$ $\Rightarrow \quad \dot{r} = -\dfrac{x^4 + y^4}{r} \le 0,$

$$\dot{\theta} = 1 - \frac{xy^3 - yx^3}{r^2} \to 1 \quad \text{as } r \to 0.$$



$H = -\frac{1}{2}(x^2 + y^2)$ for linear.

* $\begin{aligned}\dot{x} &= -y + x^3,\\ \dot{y} &= x + y^3,\end{aligned}$ $\Rightarrow \quad \dot{r} = \dfrac{x^4 + y^4}{r} \ge 0,$

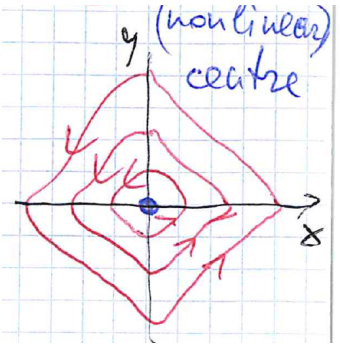
$$\dot{\theta} \to 1 \quad \text{as } r \to 0.$$



* $\begin{aligned}\dot{x} &= -y - 2x^2y,\\ \dot{y} &= x + 2y^2x\end{aligned}$ Hamiltonian system,

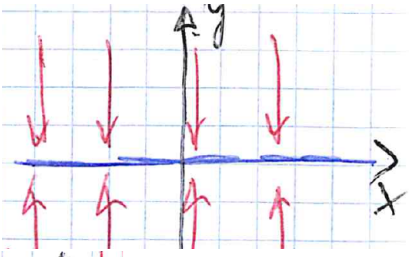
$$\dot{x} = \frac{\partial H}{\partial y}, \qquad \dot{y} = -\frac{\partial H}{\partial x},$$

with $H = -\frac{1}{2}(x^2 + y^2) - x^2y^2$ — contours are orbits.



(b) $\lambda_1 = 0$, $\lambda_2 \neq 0$, linearized — line of fixed points.

Some possibilities:



* $\begin{aligned} \dot{x} &= x^2, \\ \dot{y} &= -y, \end{aligned}$ linearized $\begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix}$,

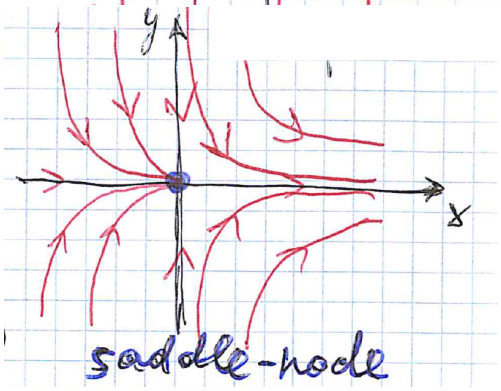
$$\text{Det}\,\hat{A} = 0, \quad \text{Tr}\,\hat{A} = -1.$$

Full solution:

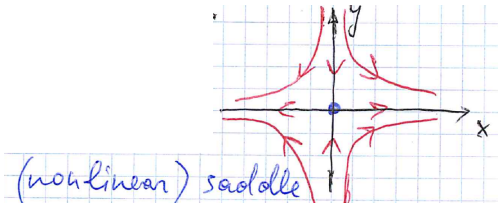
$$y \propto \exp\left(\frac{1}{x}\right), \quad \dot{x} > 0, \quad y \propto e^{-t}.$$

Note: $E^c = \{y = 0\}$.

In the nonlinear case have multiple (non-unique) manifolds tangent to E^c at $(0,0)$ — no analogue of stable manifold theorem for centre manifolds.

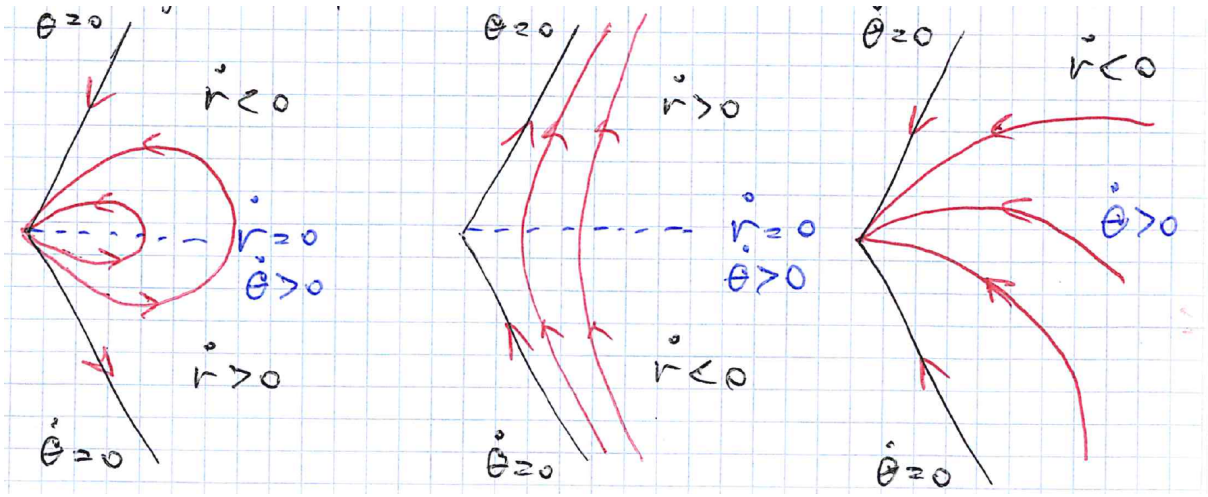


* $\begin{aligned} \dot{x} &= x^3, \\ \dot{y} &= -y, \end{aligned}$ $y \propto \exp\left(\frac{1}{2x^2}\right)$.



(c) $\lambda_1 = \lambda_2 = 0$, linearized — plane or line of fixed points.

Almost anything is possible! Change to polars, find rays $\dot{\theta} = 0$, between them can have



§2.3 Sketching phase portraits

Example $\quad \dot{x} = x(1-y), \quad \dot{y} = -y + x^2, \quad x \mapsto -x$ symmetry, $\Rightarrow$ only $x \geq 0$ is enough.

Fixed points $(0,0)$, $(1,1)$, $(-1,1)$. FP at $(0,0)$ is a saddle, studied before. Focus on $(1,1)$.

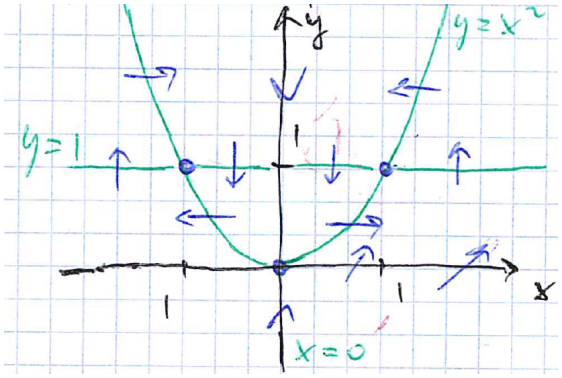
Linearize:

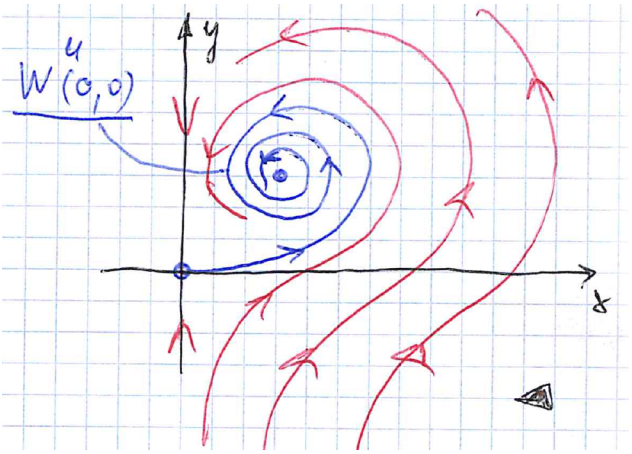
$$\hat{A} = \begin{pmatrix} 1-y & -x \\ 2x & -1 \end{pmatrix}, \qquad \hat{A}_{(1,1)} = \begin{pmatrix} 0 & -1 \\ 2 & -1 \end{pmatrix},$$

$$\operatorname{Tr}\hat{A} = -1, \quad \operatorname{Det}\hat{A} = 2, \quad (\operatorname{Tr}\hat{A})^2 < 4\operatorname{Det}\hat{A} \quad \Rightarrow \quad \text{focus},$$

$$\lambda = \rho \pm i\omega \quad \text{and} \quad \rho = \tfrac{1}{2}\operatorname{Tr}\hat{A} < 0 \quad \Rightarrow \quad \text{stable focus}.$$

Nullclines (where $\dot{x} = 0$, $\dot{y} = 0$ separately) get signs of $\dot{y}$, $\dot{x}$ on them.





General strategy

(1) Check for symmetries, find fixed points

(2) Calculate Jacobian, classify fixed points

(3) For saddles, find eigenvectors of Jacobian

(4) Find nullclines (where $\dot{x}$ or $\dot{y} = 0$) and determine flow direction on them and around

(5) Use knowledge of stable/unstable manifolds

(6) Combine these features into a global picture

(7) Trajectories cannot cross!

(8) Search for periodic orbits (see Chapter 4)

End of lecture 7

§3. Stability

If we found a fixed point, or more generally an invariant set Λ, we know that starting at Λ system will remain there. But what happens if we slightly perturb the system away from Λ? Does the system remain near Λ? Or does it converge back to or away from Λ? These are the issues of system's stability, for which there are several definitions.

§3.1 Definitions

Let Λ be an invariant set (a fixed point, periodic orbit, etc.) of a general (autonomous) system described by a flow ϕ_t.

Definition Neighborhood of a set Λ

For $\delta > 0$ the neighborhood of Λ is defined as

$$N_\delta(\Lambda) = \{\bar{x} : \exists\, \bar{y} \in \Lambda \text{ such that } |\bar{x} - \bar{y}| < \delta\}.$$

Definition Flow tending to Λ

The flow $\phi_t \to \Lambda$ if

$$\inf_{\bar{y}\in\Lambda} |\phi_t(\bar{x}) - \bar{y}| \to 0 \quad \text{as } t \to \infty.$$

Definition Lyapunov stability

The set Λ is Lyapunov stable if $\forall\, \varepsilon > 0\ \exists\, \delta > 0$ such that if $\bar{x} \in N_\delta(\Lambda)$ then $\phi_t(\bar{x}) \in N_\varepsilon(\Lambda) \quad \forall\, t \geq 0$.

"Start near, stay near"

Definition Quasi-asymptotic stability (QAS)

The set Λ is quasi-asymptotically stable if $\exists\, \delta > 0$ such that if $\bar{x} \in N_\delta(\Lambda)$ then $\phi_t(\bar{x}) \to \Lambda$ as $t \to \infty$.

"Get close eventually"

Definition Asymptotic stability

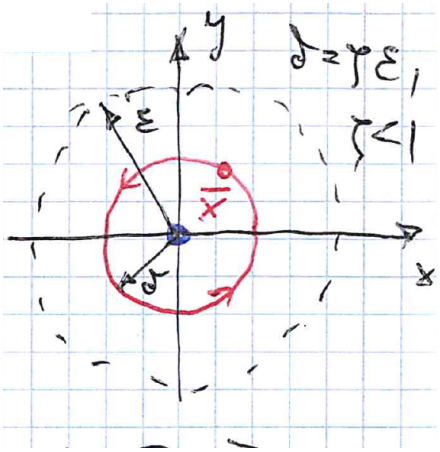
The set Λ is asymptotically stable if it is both Lyapunov stable and quasi-asymptotically stable.

"Stays near and converges to Λ"

Example Λ is LS but not QAS

$$\begin{pmatrix}\dot{x}\\ \dot{y}\end{pmatrix} = \begin{pmatrix}-y\\ x\end{pmatrix}$$

Hamiltonian system with circular orbits. For $\forall \varepsilon$ take $\delta = \zeta\varepsilon$, $0 < \zeta < 1$, then for $|\bar{x}| < \delta$ always have $|\phi_t(\bar{x})| < \varepsilon$, $\Rightarrow$ origin is LS. But no convergence to $\bar{x} = \bar{0}$, so not QAS, and not AS.

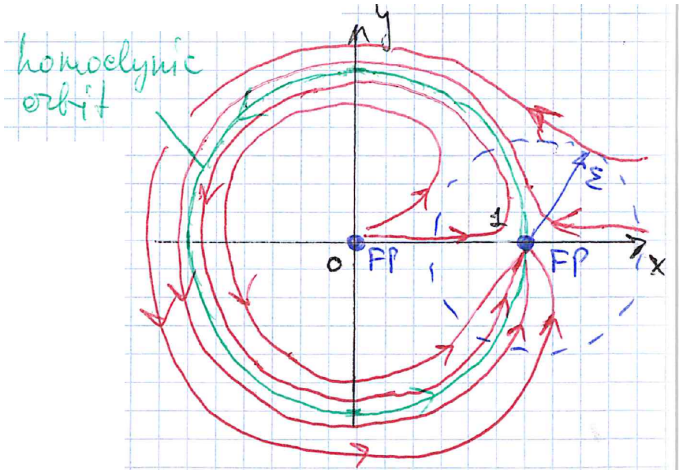


Example Λ is QAS but not LS

$$\begin{pmatrix}\dot{r}\\ \dot{\theta}\end{pmatrix} = \begin{pmatrix}r(1-r^2)\\ \sin^2\frac{\theta}{2}\end{pmatrix}$$

Linearized has one $\lambda = 0$, line of fixed points $\parallel$ to y-axis starting at $(1,0)$.

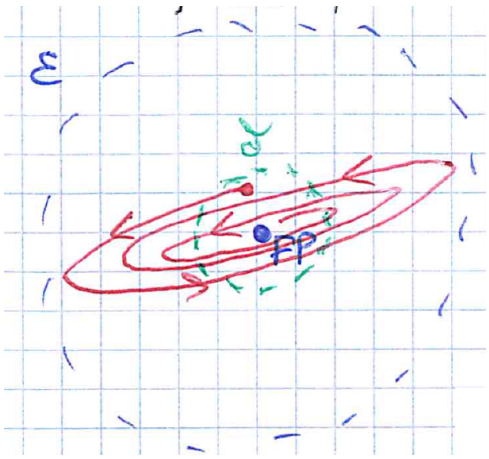
Nonlinear terms turn $(1,0)$ into a saddle-node.



For $\varepsilon < 1$ cannot find $\delta > 0$ such that trajectory is in $N_\varepsilon((1,0))$ — it will leave it. So, not LS! But is QAS as all trajectories eventually converge at FP $(1,0)$. So, not AS.

Example AS with non-monotonic convergence

Stable focus, elongated in shape.



LS since for any $\varepsilon > 0$ can always find $\delta > 0$ such that $\phi_t(\bar{x})$ with $\bar{x} \in N_\delta(\bar{0})$ always fits in $N_\varepsilon(\bar{0})$. Also QAS, obviously, and thus AS.

But approaches FP non-monotonically!

Definition Basin of attraction

If Λ is an asymptotically stable invariant set, its basin of attraction is

$$B(\Lambda) = \{\bar{x} : \phi_t(\bar{x}) \to \Lambda \text{ as } t \to \infty\}.$$

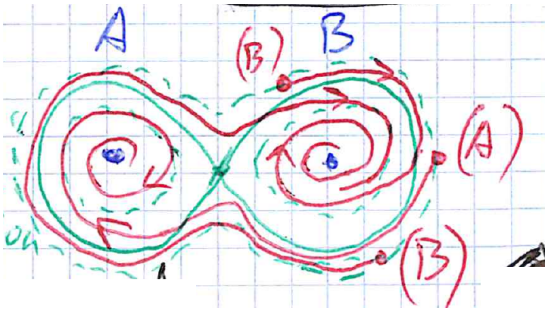
If $B(\Lambda) = \mathbb{R}^n$ then Λ is globally attracting (or globally stable).

Example In last two examples FPs were globally attracting (with some caveats in the saddle-node case).

Example Basin of stability can be non-trivial.

E.g. double potential well with weak friction.

Hamiltonian contours w/o friction



For isolated FPs can use linearized system in some cases.

Theorem Stability of hyperbolic fixed points.

If $\bar{x}_0$ is a hyperbolic stable focus or stable node then it is asymptotically stable (all hyperbolic sinks are AS). If $\bar{x}_0$ is a hyperbolic fixed point with at least one $\mathrm{Re}(\lambda_i) > 0$, then it is unstable.

§3.2 Lyapunov functions

We can prove important results about the stability of fixed points of a given flow $\phi_t(\bar{x})$ if we can find some functions that behave in a certain way under the flow.

Definition A continuously differentiable (C^1) function $V(\bar{x}) : \mathbb{R}^n \to \mathbb{R}$ is a Lyapunov function for a flow $\phi_t(\bar{x})$ on a domain D containing the origin (fixed point) if

(i) $V(\bar{0}) = 0$ and $V(\bar{x}) > 0$ for $\bar{x} \neq \bar{0}$ in D and

(ii) $\dot{V} = \bar{f} \cdot \nabla V \leq 0$ in D (or $V(\phi_t(\bar{x})) \leq V(\bar{x})$ for $\forall\, t \geq 0$).

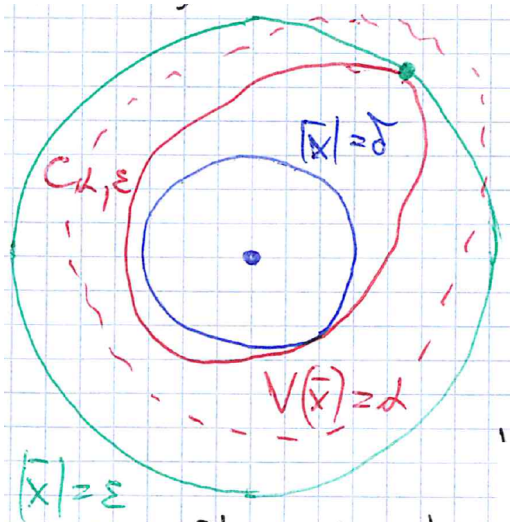
Then we have these 2 theorems.

Theorem Lyapunov's 1st theorem

Suppose that a dynamical system $\dot{\bar{x}} = \bar{f}(\bar{x})$ has a fixed point at the origin. If a Lyapunov function exists, then the origin is Lyapunov stable.

Proof Take $\varepsilon > 0$ such that $\{|\bar{x}| < \varepsilon\} \subseteq D$. Let

$$\alpha = \min\{V(\bar{x}) : |\bar{x}| = \varepsilon\}.$$



Choose $\delta > 0$ such that

$$\{|\bar{x}| < \delta\} \subset C_{\alpha,\varepsilon}$$

$$= \{\bar{x} : V(\bar{x}) < \alpha,\ |\bar{x}| < \varepsilon\},$$

Then if $|\bar{x}| < \delta$,

$$V(\phi_t(\bar{x})) \leq V(\bar{x}) < \alpha,$$

$$\Rightarrow \quad \underline{|\phi_t(\bar{x})| < \varepsilon}.$$

So the origin is Lyapunov stable (LS).

Intuitively, to leave the ball $|\bar{x}| < \varepsilon$, a trajectory must first cross the boundary $|\bar{x}| = \varepsilon$, where $V \geq \alpha$ by the definition of α. However, V starts below α and is non-increasing along the trajectory, so it always remains below α. The trajectory therefore cannot reach the boundary and cannot leave the ball.

End of Lecture 8

Theorem Lyapunov's 2nd theorem

If, in addition, $\dot{V}(\bar{x}) < 0$ for $\bar{x} \neq \bar{0}$ in D, then V is a strict Lyapunov function and the origin is asymptotically stable.

Proof By the first theorem, trajectories starting sufficiently near the origin stay in a compact neighbourhood $K \subset D$. For such an initial point $\bar{x}_0$, $V(\phi_t(\bar{x}_0))$ decreases and is bounded below by zero, so

$$V(\phi_t(\bar{x}_0)) \longrightarrow V_0 \geq 0.$$

Suppose $V_0 > 0$. By continuity at the origin, the trajectory is bounded away from $\bar{0}$. On the resulting compact subset of K, strict negativity and continuity of $\dot{V}$ give a constant $\beta > 0$ such that

$$\dot{V}(\phi_t(\bar{x}_0)) \leq -\beta < 0.$$

But then

$$V(\phi_t(\bar{x}_0)) = V(\bar{x}_0) + \int_0^t \dot{V}(\phi_s(\bar{x}_0))\, ds \leq V(\bar{x}_0) - \beta t,$$

which eventually becomes negative, a contradiction. Thus $V_0 = 0$. Since V is positive away from $\bar{0}$ and the trajectory stays in K, $\phi_t(\bar{x}_0) \to \bar{0}$. Together with Lyapunov stability, this proves AS.

Example Stability of the origin

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \begin{pmatrix} -2x \\ x - y \end{pmatrix}.$$

This is a linear system with Jacobian

$$A = \begin{pmatrix} -2 & 0 \\ 1 & -1 \end{pmatrix}, \qquad \lambda_1 = -1, \quad \lambda_2 = -2,$$

so the origin is a stable node. Let us illustrate stability using a Lyapunov function.

Try $V(x, y) = ax^2 + by^2$, where $a, b > 0$. Clearly $V \geq 0$ and $V = 0$ only at the origin. Compute

$$\dot{V} = V_x \dot{x} + V_y \dot{y} = 2ax(-2x) + 2by(x - y) = -2b\left(\frac{2a}{b}x^2 - xy + y^2\right).$$

(i) Take $b = 8a$. Then

$$\dot{V} = -2b\left(\frac{x}{2} - y\right)^2 \leq 0.$$

This is a Lyapunov function, but is not strict, since $\dot{V} = 0$ on $\{y = x/2\}$, which extends beyond the origin. Still, Lyapunov's first theorem establishes Lyapunov stability (LS).

(ii) Take $b = 2a$. Then

$$\dot{V} = -2b\left[\frac{3}{4}x^2 + \left(\frac{x}{2} - y\right)^2\right] < 0 \qquad \text{for } (x, y) \neq (0, 0).$$

A strict Lyapunov function has been found, so the second theorem shows that the origin is asymptotically stable (AS).

Even when the conditions of the second theorem are not fulfilled, we can sometimes demonstrate asymptotic stability of the origin.

Theorem LaSalle's Invariance Principle

Let D be a compact (closed and bounded), positively invariant set, and let V be continuously differentiable on a neighbourhood of D, with $\dot{V} \leq 0$ in D. Then for each $\bar{x} \in D$ the orbit is bounded, and there exists c such that

$$\omega(\bar{x}) \subseteq M_c = \{\bar{z} \in D : V(\phi_t(\bar{z})) = c \text{ for all } t \geq 0\}.$$

Equivalently, $\phi_t(\bar{x})$ tends to an invariant subset of

$$E = \{\bar{z} \in D : \dot{V}(\bar{z}) = 0\}.$$

Proof Choose $\bar{x} \in D$ and let

$$c = \inf_{t \geq 0} V(\phi_t(\bar{x})).$$

Since D is compact and positively invariant, the ω-limit set exists. Take $\bar{y} \in \omega(\bar{x})$. By definition, there is a sequence $t_n \to \infty$ such that $\phi_{t_n}(\bar{x}) \to \bar{y}$. Since V is continuous and non-increasing along orbits,

$$V(\bar{y}) = \lim_{n \to \infty} V(\phi_{t_n}(\bar{x})) = c.$$

Now show that $V(\phi_t(\bar{y})) = c$. Suppose instead that $V(\phi_T(\bar{y})) < c$ for some $T \geq 0$. By continuity of the flow,

$$V(\phi_{t_n+T}(\bar{x})) = V(\phi_T(\phi_{t_n}(\bar{x}))) \longrightarrow V(\phi_T(\bar{y})) < c.$$

Thus there would be a time $\tau = t_n + T$ with $V(\phi_\tau(\bar{x})) < c$, contradicting the definition of c. Hence $V(\phi_t(\bar{y})) = c$ for all $t \geq 0$. Every point of $\omega(\bar{x})$ therefore lies on an orbit on which V is constant; this gives the invariant subset of E claimed above.

Notes

- The ω-limit set need not be a fixed point. This is a more general statement than Lyapunov's first or second theorem. In practice, we will use LaSalle's principle to determine the stability of fixed points.
- If V is positive definite near the origin and the largest invariant subset of $\{\bar{z} : \dot{V}(\bar{z}) = 0\}$ in a suitable compact neighbourhood is $\{\bar{0}\}$, then the origin is asymptotically stable. This strengthens the conclusion of the first theorem without requiring a strict Lyapunov function.

Example Explore stability of the origin

$$\begin{pmatrix} \dot{x} \\ \dot{y} \end{pmatrix} = \begin{pmatrix} y - xy^2 \\ -x^3 \end{pmatrix}.$$

Without the $-xy^2$ term, this system would be Hamiltonian, with

$$H(x, y) = \frac{1}{2}y^2 + \frac{1}{4}x^4.$$

Take $V(x, y) = H(x, y)$. Then

$$\dot{V} = H_x\dot{x} + H_y\dot{y} = x^3(y - xy^2) + y(-x^3) = -x^4y^2 \leq 0.$$

Thus H is a Lyapunov function on $\mathbb{R}^2$. It is not strict, since

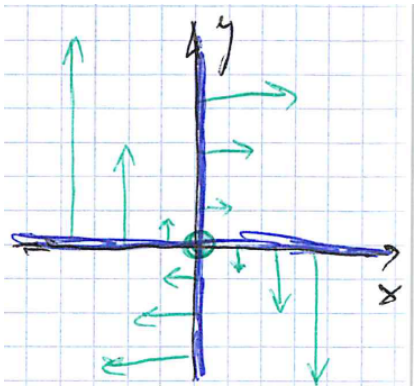
$$\dot{V} = 0 \quad \text{on } \{x = 0\} \cup \{y = 0\}.$$

The zero-derivative set extends beyond the origin, so the second theorem cannot be applied directly. But none of its nonzero points remains in the set:

$$x = 0,\ y \neq 0 \ \Rightarrow\ \dot{x} = y \neq 0,$$

$$y = 0,\ x \neq 0 \ \Rightarrow\ \dot{y} = -x^3 \neq 0.$$

Thus the largest invariant subset of $\{\dot{V} = 0\}$ is $\{\bar{0}\}$. By LaSalle's principle, the origin is asymptotically stable.



To apply the compact-set version of the theorem, take any sublevel set $\{H \leq k\}$, $k > 0$. It is compact and positively invariant since $\dot{H} \leq 0$.

Estimating a basin of attraction

We can use Lyapunov's theorems and LaSalle's principle to establish a basin of attraction (domain of stability, DoS) of a fixed point. Here is an algorithm:

1. Choose a function $V(\bar{x}) : \mathbb{R}^n \to \mathbb{R}$ and a domain D such that

   $$V(\bar{x}) \geq 0 \text{ in } D, \qquad V(\bar{x}) = 0 \text{ only for } \bar{x} = \bar{0},$$

   and $\dot{V}(\bar{x}) \leq 0$ in D. The latter condition may limit D.

2. Find $k > 0$ such that the sublevel set

   $$C_k = \{\bar{x} : V(\bar{x}) < k\} \subseteq D.$$

   For the compact-set theorem, choose a sublevel region around the origin whose closure is compact and contained in D.

3. Adjust k (or V) so that the largest invariant subset of

   $$\overline{C_k} \cap \{\bar{z} : \dot{V}(\bar{z}) = 0\}$$

   is the origin. LaSalle's principle then gives

   $$C_k \subseteq \text{DoS}(\bar{0}).$$

4. Take different functions V to obtain other such regions C_k. Their union forms a better (larger) estimate of $\text{DoS}(\bar{0})$.

End of lecture 9

Example Estimating the domain of stability

$$\dot{x} = x(y^2 - 1), \qquad \dot{y} = y(x^2 - 1).$$

The fixed points are $(0,0)$ and $(\pm 1, \pm 1)$. The Jacobian is

$$A(x,y) = \begin{pmatrix} y^2 - 1 & 2xy \\ 2xy & x^2 - 1 \end{pmatrix}.$$

At the origin, $A = -I$, so the origin is a stable star node. At each of the other four fixed points the eigenvalues are 2 and -2, so they are saddles. The system is symmetric under $(x,y) \mapsto (y,x)$, $x \mapsto -x$, and $y \mapsto -y$.

(1) Try the Lyapunov function

$$V(x,y) = x^2 + a^2 y^2, \qquad a > 0.$$

Then

$$\dot{V} = 2x^2(y^2 - 1) + 2a^2 y^2 (x^2 - 1) = -2(x^2 + a^2 y^2) + 2(1 + a^2) x^2 y^2.$$

We need a domain D on which $\dot{V} < 0$ away from the origin. For example, the region where this holds is described by

$$D = \left\{ (x,y) : |x| > \frac{a}{\sqrt{1+a^2}}, \quad y^2 < \frac{x^2}{(1+a^2)x^2 - a^2} \right\} \cup \left\{ (x,y) : |x| \le \frac{a}{\sqrt{1+a^2}} \right\}.$$

Here $\dot{V} < 0$ on $D \setminus \{(0,0)\}$.

(2) Find k so that $C_k = \{(x,y) : V(x,y) < k\} \subset D$. Parametrise a contour of V by

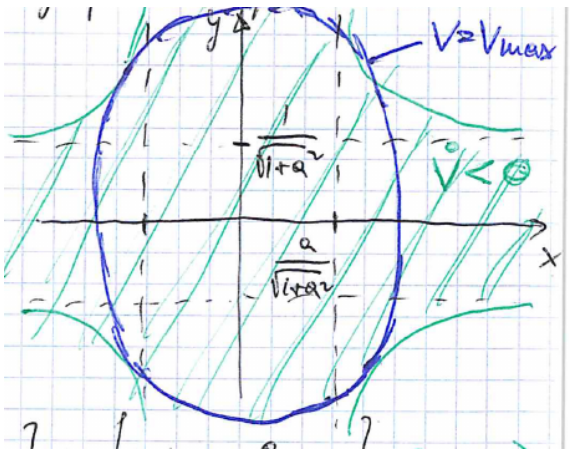
$$x = \sqrt{V} \cos\phi, \qquad y = \frac{\sqrt{V}}{a} \sin\phi.$$

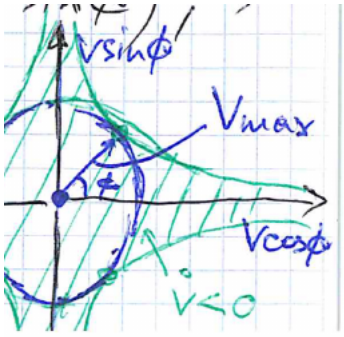
Substitution gives

$$\dot{V} = -2V + \frac{(1+a^2)V^2}{2a^2} \sin^2(2\phi) < 0 \quad \text{if} \quad 0 < V < \frac{4a^2}{(1+a^2)\sin^2(2\phi)}.$$

Thus negativity is guaranteed uniformly in ϕ whenever

$$0 < V < V_{\max} = \frac{4a^2}{1+a^2}.$$





(3) Hence

$$C_a = \left\{(x,y) : x^2 + a^2y^2 < \frac{4a^2}{1+a^2}\right\} \subseteq \mathrm{DoS}(\bar{0}).$$

LaSalle's principle is not needed here: V is strict in C_a away from the origin, so Lyapunov's second theorem applies. More precisely, every point of C_a lies in a smaller compact sublevel set on which the theorem applies.

(4) Improve the estimate by varying a. The ellipses become thin in the x direction as $a \to 0$, are the disc $x^2 + y^2 < 2$ for $a = 1$, and become thin in the y direction as $a \to \infty$:

$$\frac{x^2}{a^2} + y^2 < \frac{4}{1+a^2}, \qquad x^2 + \left(\frac{y}{a^{-1}}\right)^2 < \frac{4}{1+a^{-2}}.$$

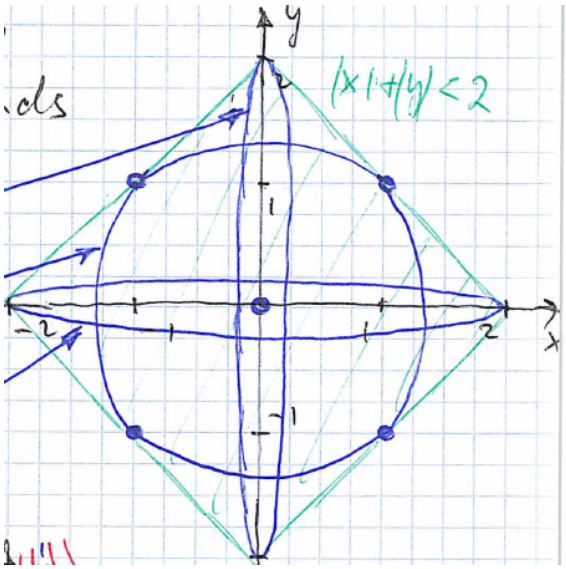
Their union is

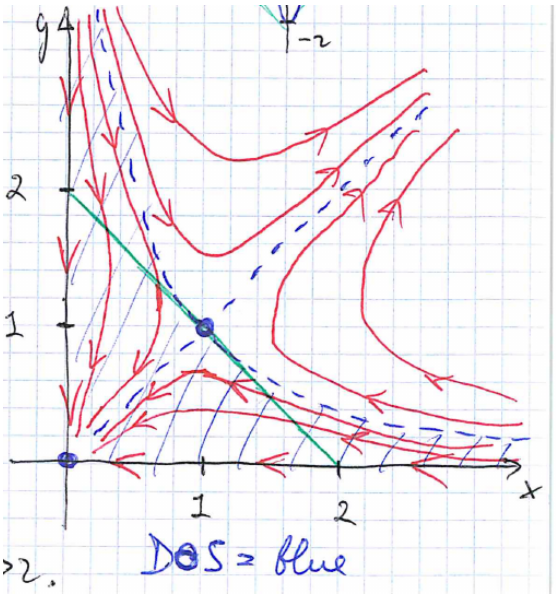
$$\bigcup_{a>0} C_a = \{(x,y) : |x| + |y| < 2\}.$$

Indeed, writing $s = a^2 > 0$, the ellipse inequality becomes

$$x^2 + y^2 + \frac{x^2}{s} + sy^2 < 4,$$

and $\inf_{s>0}(x^2/s + sy^2) = 2|xy|$. This diamond is the best estimate obtained from this family of Lyapunov functions. It is not the full domain of stability: on either coordinate axis, solutions tend to the origin even when $|x| > 2$ or $|y| > 2$.





3.3 Bounding functions

Another use for functions decreasing along trajectories is to prove that trajectories eventually enter a bounded region.

Definition Bounding function

Suppose V has bounded sublevel sets and there are constants M and $\delta > 0$ such that

$$\dot{V}(\bar{x}) \leq -\delta < 0 \qquad \text{whenever } V(\bar{x}) \geq M.$$

Then all trajectories eventually enter and remain in $\{\bar{x} : V(\bar{x}) \leq M\}$. Such a function is called a bounding function. While $V \geq M$, its value decreases at least at rate δ, giving finite entry time.

Example Consider the polar system

$$\dot{r} = r - r^3(1 + k\sin\theta) - r^5, \qquad \dot{\theta} = 1, \qquad 0 < k < 1.$$

Take $V = \frac{1}{2}r^2$. Then

$$\dot{V} = r\dot{r} = r^2\big[1 - r^2(1 + k\sin\theta) - r^4\big] \leq r^2\big[1 - r^2(1-k) - r^4\big].$$

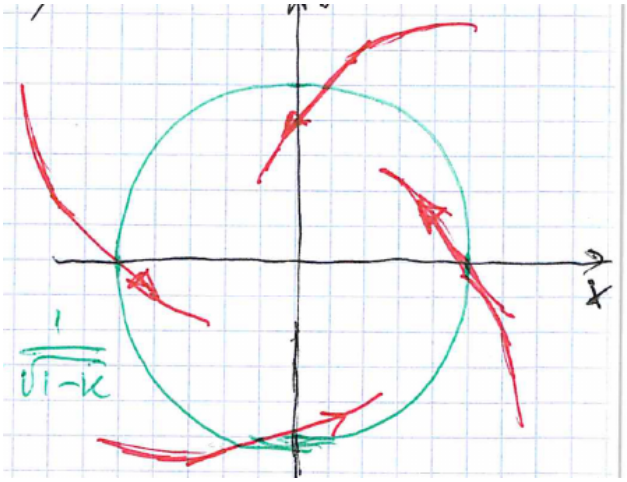
For $r^2 \geq 1/(1-k)$,

$$\dot{V} \leq -r^6 \leq -\frac{1}{(1-k)^3} < 0.$$

Thus every trajectory enters the disc

$$r \leq \frac{1}{\sqrt{1-k}}$$

and remains there. What happens within this bounded domain, where an ω-limit set exists? Is the limit a fixed point or a periodic orbit?



4. Periodic orbits in $\mathbb{R}^2$: existence and stability

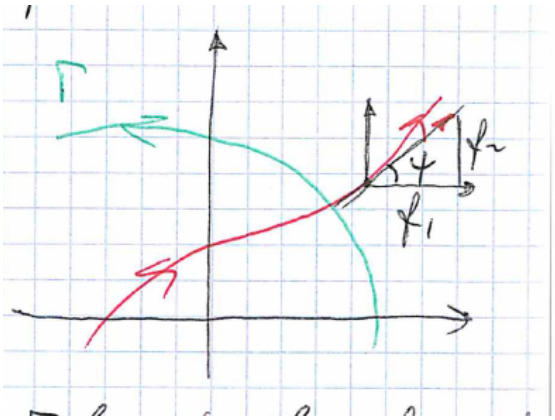
We first establish techniques for confirming or excluding the existence of periodic orbits.

4.1 Poincaré index

Let

$$\dot{x} = f_1(x, y), \qquad \dot{y} = f_2(x, y), \qquad \bar{f} = (f_1, f_2).$$

Away from fixed points, the vector field makes an angle ψ with the x axis, with $\tan\psi = f_2/f_1$ (using a continuous choice of angle). Let Γ be a simple closed curve containing no fixed point on the curve; it need not be a trajectory. Traverse Γ anticlockwise and measure the net change in ψ. After one circuit the angle changes by an integer multiple of 2π. This integer is the Poincaré index $I(\Gamma)$.



Its explicit form is

$$I(\Gamma) = \frac{1}{2\pi}\oint_\Gamma d\psi = \frac{1}{2\pi}\oint_\Gamma \frac{f_1\,df_2 - f_2\,df_1}{f_1^2 + f_2^2}.$$

Any parametrisation of Γ may be used to compute this integral. For example, $df_2 = f_{2,x}\,dx + f_{2,y}\,dy$. On a circle,

$$x = R\cos\phi, \quad y = R\sin\phi, \qquad df_2 = R(-f_{2,x}\sin\phi + f_{2,y}\cos\phi)\,d\phi.$$

At a fixed point $f_1 = f_2 = 0$, the angle is undefined; a curve used to compute $I(\Gamma)$ must avoid such points.

Properties

1. $I(\Gamma)$ is unchanged by a continuous deformation of Γ as long as no fixed point is crossed. Indeed, the integral varies continuously under small deformations, whereas the index is integer-valued, so it must stay constant.

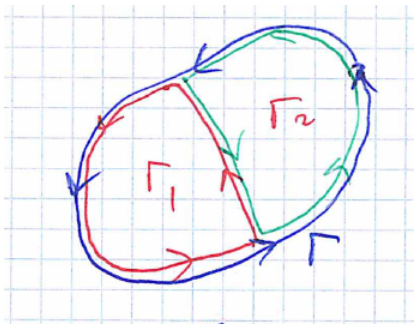
2. If Γ encloses no fixed points, then $I(\Gamma) = 0$. Shrink it to a small contour on which the field is nearly uniform, with zero net change of angle.

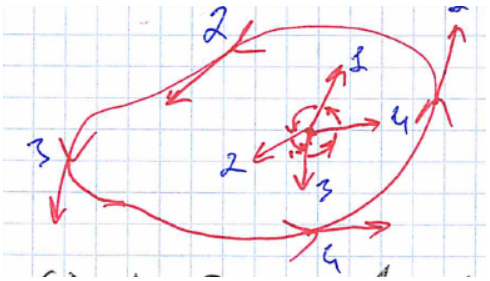
3. The index is additive: if a contour is split into Γ_1 and Γ_2, with the shared joins traversed in opposite directions, then
$$I(\Gamma) = I(\Gamma_1) + I(\Gamma_2).$$
The integrals on the shared joins cancel.

4. Reversing time does not change $I(\Gamma)$: replacing $\bar{f}$ by $-\bar{f}$ adds π to the angle but does not change its net rotation.

5. For a closed trajectory (a periodic orbit), $I(\Gamma) = +1$, since the vector field is tangent to the orbit at every point and makes one positive turn as the curve is traversed anticlockwise.

6. The index of an isolated fixed point is the index of any sufficiently small simple closed curve encircling that fixed point alone.





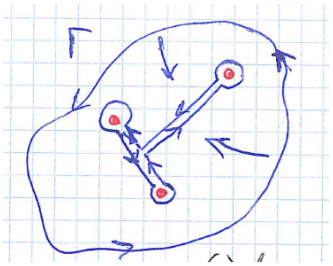


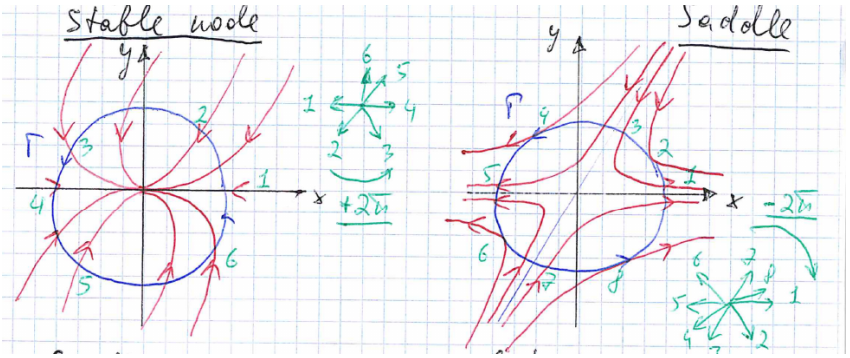
End of lecture 10

Poincaré index: further properties

7. For a curve encircling finitely many isolated fixed points, its index is the sum of the indices of those fixed points. This follows by deforming the curve into small loops around the individual points and using additivity.

8. A hyperbolic sink or source has index $+1$, while a hyperbolic saddle has index -1. Non-hyperbolic fixed points must be treated case by case; for example, a saddle-node has index 0.



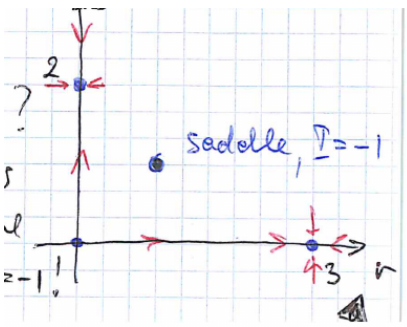


Corollary Any periodic orbit must encircle one or more fixed points, and the sum of their indices must be $+1$. This can be used to exclude periodic orbits.

Example Lotka–Volterra system

$$\dot{r} = r(3 - r - 2s), \qquad \dot{s} = s(2 - r - s).$$

The fixed points $(0,0)$, $(3,0)$ and $(0,2)$ are nodes, and $(1,1)$ is a saddle of index -1. A periodic orbit with $r, s > 0$ cannot cross the invariant coordinate axes, so it cannot encircle any of the three nodes. It can encircle either no fixed point, giving index 0, or only the saddle, giving index -1. Neither is compatible with the index $+1$ of a periodic orbit. Thus no such periodic orbit exists.



4.2 Dulac's criterion

Theorem Dulac's theorem

If there is a continuously differentiable function $\varphi(x, y)$ such that

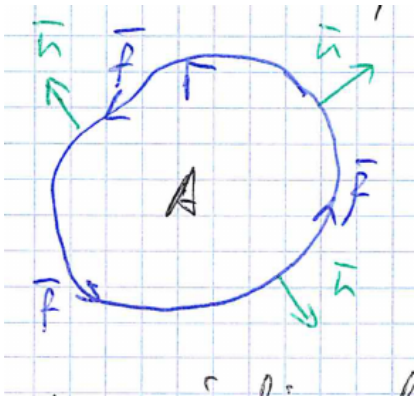
$$\nabla \cdot (\varphi \bar{f}) > 0 \qquad \text{everywhere in } D,$$

or $\nabla \cdot (\varphi \bar{f}) < 0$ everywhere in D, where $D \subset \mathbb{R}^2$ is a simply connected domain, then no periodic orbit lies entirely in D.

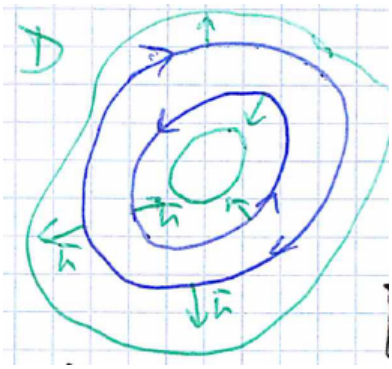
Proof Suppose $\nabla \cdot (\varphi \bar{f}) > 0$ and a periodic orbit Γ exists. Let A be its interior and $\hat{n}$ the outward unit normal. The field is tangent to Γ, so the divergence theorem gives

$$0 = \oint_\Gamma \varphi \bar{f} \cdot \hat{n} \, d\ell = \iint_A \nabla \cdot (\varphi \bar{f}) \, dA > 0,$$

a contradiction. The other sign is treated similarly.



- Taking $\varphi = 1$ gives the divergence test: if $\nabla \cdot \bar{f}$ is strictly positive or strictly negative throughout a simply connected domain, there is no periodic orbit in that domain.
- If $\nabla \cdot (\varphi \bar{f})$ has a strict constant sign in a doubly connected domain, there is at most one periodic orbit, and it must encircle the hole. Otherwise, applying the divergence theorem to the region between two periodic orbits gives the same contradiction.



Example Lotka–Volterra system

$$\dot{r} = r(a - br - cs), \qquad \dot{s} = s(d - er - fs), \qquad b, f > 0.$$

For $r, s > 0$, take $\varphi = 1/(rs)$. Then

$$\nabla \cdot (\varphi \bar{f}) = \frac{\partial}{\partial r}\left(\frac{a - br - cs}{s}\right) + \frac{\partial}{\partial s}\left(\frac{d - er - fs}{r}\right) = -\frac{b}{s} - \frac{f}{r} < 0.$$

There are no periodic orbits in the positive quadrant. In the preceding example, $b = f = 1$.

The divergence theorem extends to $\mathbb{R}^n$ and can exclude closed invariant hypersurfaces enclosing nonzero volume when the divergence has a strict constant sign throughout the enclosed region.

Example Lorenz system

$$\dot{x} = \sigma(y - x), \qquad \dot{y} = rx - y - zx, \qquad \dot{z} = -bz + xy, \qquad r, \sigma, b > 0.$$

Here

$$\nabla \cdot \bar{f} = -\sigma - 1 - b < 0.$$

Hence there is no closed invariant surface of dimension 2 enclosing a nonzero volume. This does not exclude one-dimensional periodic orbits.

Theorem Gradient criterion

Suppose there is a positive function $\rho(x, y)$ such that

$$\rho \bar{f} = \nabla \psi$$

for a single-valued potential ψ on a simply connected domain D. Then there are no periodic orbits in D.

Proof On a hypothetical nonconstant periodic orbit Γ, traversed in the direction of the flow,

$$0 < \oint_\Gamma \rho \bar{f} \cdot d\bar{\ell} = \oint_\Gamma \nabla \psi \cdot d\bar{\ell} = 0.$$

The last integral vanishes because the potential is single-valued. This contradiction excludes periodic orbits.

Example Does the system

$$\dot{x} = 2x + x^2 y - y^3, \qquad \dot{y} = -2y - xy^2 + x^3$$

have periodic orbits? Notice that

$$e^{xy} \bar{f} = \nabla \left[e^{xy} (x^2 - y^2) \right].$$

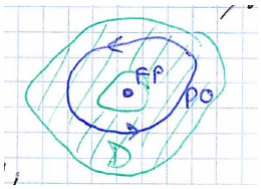
Since $\rho = e^{xy} > 0$ and $\psi = e^{xy}(x^2 - y^2)$ is single-valued on $\mathbb{R}^2$, the gradient criterion excludes periodic orbits.

4.3 Poincaré–Bendixson theorem

Theorem Poincaré–Bendixson

If the forward orbit $O^+(\bar{x})$ of some point $\bar{x}$ remains in a compact set $D \subset \mathbb{R}^2$ containing no fixed points, then $\omega(\bar{x})$ is a periodic orbit. A sketch proof is given below.

- This result is specific to $\mathbb{R}^2$; there is no corresponding general statement in higher dimensions.
- Such a trapping region must leave a hole containing a fixed point: a periodic orbit has index $+1$. In particular, a simply connected domain containing the whole interior of that orbit cannot be free of fixed points.



Example Consider

$$\dot{x} = x - y - 2x(x^2 + y^2), \qquad \dot{y} = y + x - y(x^2 + y^2).$$

Linearising at the origin gives

$$A = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}, \qquad \det A = 2, \quad \operatorname{tr} A = 2,$$

so the origin is an unstable focus. Far from the origin the cubic terms push trajectories inwards, suggesting a periodic orbit. In polar coordinates,

$$\dot{r} = \frac{x\dot{x} + y\dot{y}}{r} = r\big[1 - r^2(1 + \cos^2\theta)\big], \qquad \dot{\theta} = 1 + \frac{1}{2}r^2 \sin(2\theta).$$

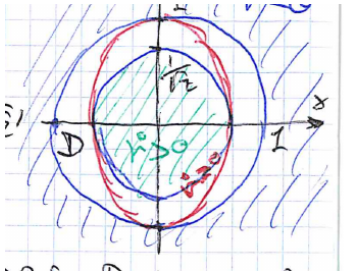
The curve $\dot{r} = 0$ away from the origin is

$$r = \frac{1}{\sqrt{1 + \cos^2\theta}}, \qquad 2x^2 + y^2 = 1.$$

We have $\dot{r} < 0$ for $r > 1$ and $\dot{r} > 0$ for $0 < r < 1/\sqrt{2}$. Thus the compact annulus

$$D = \{(x, y) : 1/\sqrt{2} \le r \le 1\}$$

is positively invariant. Moreover $\dot{\theta} \ge 1/2 > 0$ on D, so there are no fixed points in D. The Poincaré–Bendixson theorem implies the existence of a periodic orbit there.



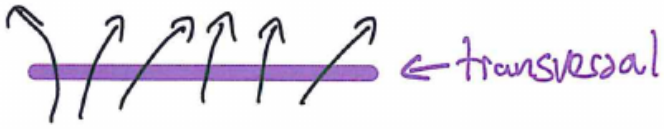
Bonus: uniqueness. In D,

$$\nabla \cdot \bar{f} = 2 - 2x^2 - 5r^2 \le -\frac{1}{2} - 2x^2 < 0.$$

By the divergence test for a doubly connected region, there is at most one periodic orbit in D. Hence the periodic orbit just obtained is unique.

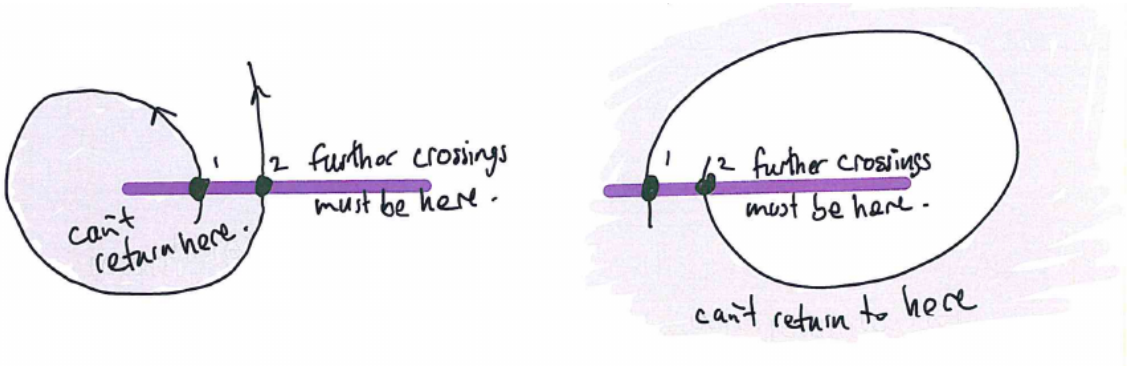
Sketch proof of the Poincaré–Bendixson theorem

The key idea is a transversal: a curve segment that nearby trajectories cross in the same direction. A short segment perpendicular to the flow can be placed through any non-fixed point.



Lemma If a nonperiodic trajectory meets a transversal several times, its successive intersections are arranged monotonically along the transversal.

After two intersections, the intervening trajectory segment and the transversal bound a region. Uniqueness of trajectories and the Jordan curve theorem restrict all further crossings to one direction along the transversal. This separation into an inside and an outside is where planarity is essential.

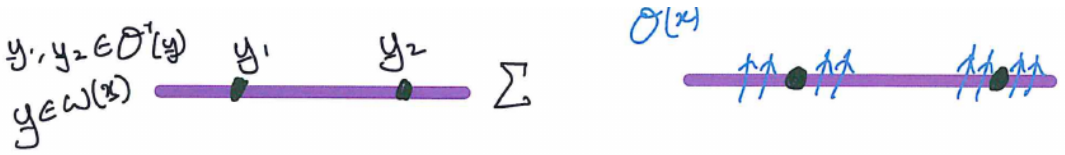


Proof sketch

1. Choose $y \in \omega(x)$, then $z \in \omega(y)$. Compactness makes these ω-limit sets nonempty, and invariance ensures they remain in D, where there are no fixed points. We aim to show that y is periodic.

2. Choose a transversal Σ through z. Since $z \in \omega(y)$, the forward orbit of y comes arbitrarily close to z. By following these nearby points forwards or backwards for a small time, obtain two intersections $y_1, y_2 \in O^+(y) \cap \Sigma$ at distinct times.

3. Both y_1 and y_2 lie in $\omega(x)$ by invariance. The same local transversal argument gives subsequences of intersections of $O^+(x)$ with Σ tending to y_1 and to y_2. If $O^+(x)$ is nonperiodic, the lemma makes its successive intersections monotone, so they cannot have two distinct limits. Hence $y_1 = y_2$, giving a return to the same point at distinct times; y therefore lies on a periodic orbit. If $O^+(x)$ is already periodic, the conclusion is immediate.



This establishes that $\omega(x)$ contains a periodic orbit. To obtain the full statement, one also checks that $\omega(x)$ meets any local transversal in at most one point. Together with connectedness of ω-limit sets of precompact forward orbits, this shows that $\omega(x)$ is exactly that one periodic orbit. The handout leaves these final details to the reader.

End of lecture 11

4.4 Near-Hamiltonian flows and periodic orbits

Consider a perturbation of a planar Hamiltonian system:

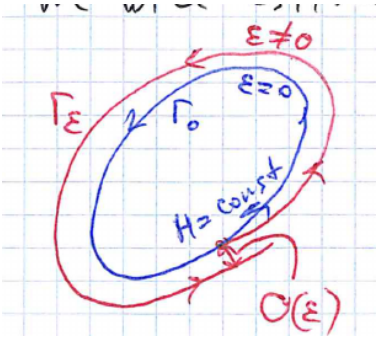
$$\begin{pmatrix}\dot{x}\\ \dot{y}\end{pmatrix} = \begin{pmatrix}f_1(x,y)+\varepsilon g_1(x,y)\\ f_2(x,y)+\varepsilon g_2(x,y)\end{pmatrix}, \qquad f_1 = H_y, \quad f_2 = -H_x.$$

- For $\varepsilon = 0$, $\dot{H} = 0$: trajectories lie on contours of H, and regular closed contours give periodic orbits.
- For $\varepsilon \neq 0$,
$$\dot{H} = H_x\dot{x} + H_y\dot{y} = \varepsilon(f_1 g_2 - f_2 g_1).$$
If a nonconstant periodic orbit Γ of period T exists, then
$$0 = \int_0^T \dot{H}\,dt = \varepsilon \int_0^T (f_1 g_2 - f_2 g_1)\,dt.$$
Thus no periodic orbit can lie wholly in a region where $f_1 g_2 - f_2 g_1$ has a strict constant sign.
- For $0 < \varepsilon \ll 1$, $\dot{H} = O(\varepsilon)$, so trajectories are weakly perturbed over finite times. The family of Hamiltonian periodic orbits need not survive. The energy balance method tests which orbits can persist.

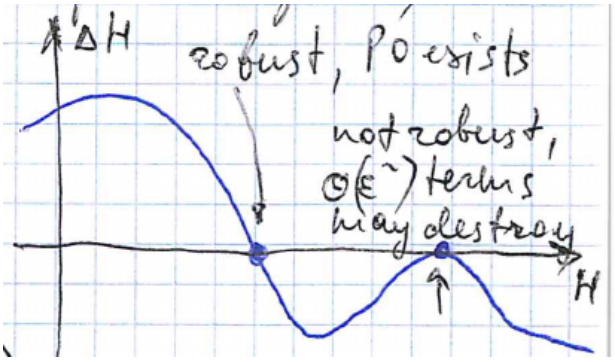
Let Γ_0 be a regular closed orbit with $H = H_0$ and period T_0. Follow a nearby perturbed trajectory Γ_ε for one return to a local transversal, taking time T_ε. Away from a separatrix, smooth dependence gives $\Gamma_\varepsilon = \Gamma_0 + O(\varepsilon)$ and $T_\varepsilon = T_0 + O(\varepsilon)$. Hence

$$\begin{aligned}\Delta H(\Gamma_\varepsilon) &= \varepsilon \int_0^{T_\varepsilon} (f_1 g_2 - f_2 g_1)(\Gamma_\varepsilon(t))\,dt\\ &= \varepsilon M(H_0) + O(\varepsilon^2),\\ M(H_0) &= \int_0^{T_0} (f_1 g_2 - f_2 g_1)(\Gamma_0(t))\,dt.\end{aligned}$$

Search for $M(H_0) = 0$. A simple zero is robust and gives a nearby periodic orbit for sufficiently small ε, by the local return map and the implicit function theorem. A double zero is inconclusive: the neglected $O(\varepsilon^2)$ terms may remove it.



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Example Consider

$$\dot{x} = y, \qquad \dot{y} = -x + x^2 + \varepsilon y(a - x),$$

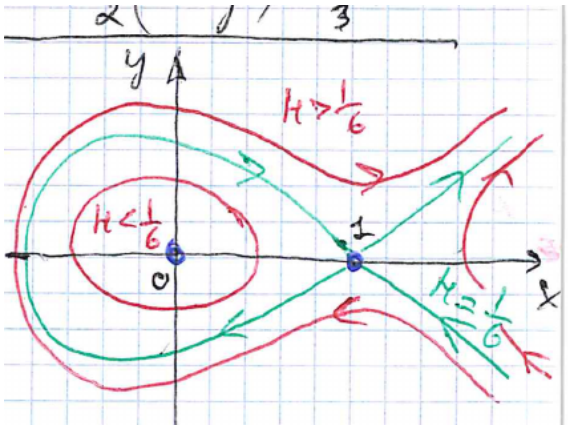
or, equivalently,

$$\ddot{x} + x - x^2 - \varepsilon \dot{x}(a - x) = 0.$$

For $\varepsilon = 0$ the Hamiltonian is

$$H(x, y) = \frac{1}{2}(x^2 + y^2) - \frac{1}{3}x^3.$$

The origin is a centre, $(1, 0)$ is a saddle, and the bounded components of $H = H_0$ for $0 < H_0 < 1/6$ are periodic orbits. The level $H = 1/6$ contains a homoclinic orbit to the saddle (a separatrix).



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For $\varepsilon \neq 0$, $g_1 = 0$, $g_2 = y(a - x)$, so

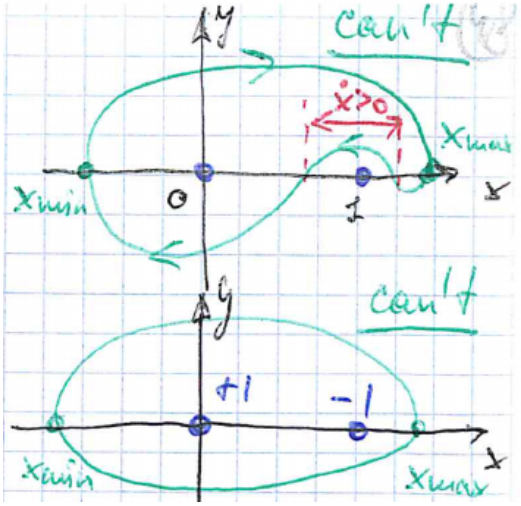
$$\dot{H} = \varepsilon y^2(a - x), \qquad \Delta H = \varepsilon \int y^2(a - x)\, dt.$$

The fixed points remain $(0, 0)$ and $(1, 0)$, with indices $+1$ and -1. Every periodic orbit must enclose the origin and exclude the saddle. Moreover, $\dot{H}$ must change sign along it, so it crosses $x = a$.

At extrema of x, $\dot{x} = y = 0$ and $\ddot{x} = x(x - 1)$. At a maximum, $0 < x_{\max} < 1$; at a minimum, $x_{\min} < 0$. Equality would put the trajectory at a fixed point. Consequently,

$$x_{\min} < a < x_{\max} < 1.$$

In particular, there are no periodic orbits for $a \geq 1$.



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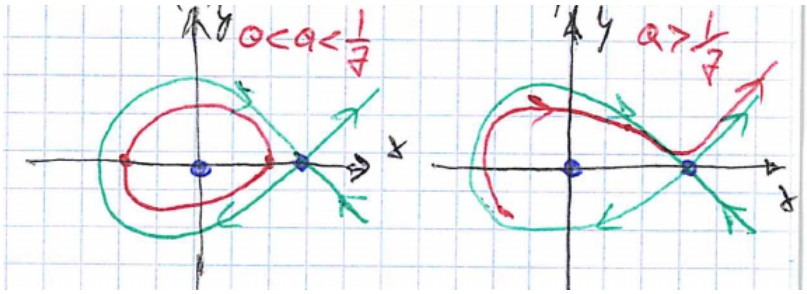
Energy balance for $0 < \varepsilon \ll 1$

On the unperturbed orbit Γ_0, $y\,dt = dx$. Symmetry in y gives the leading energy change

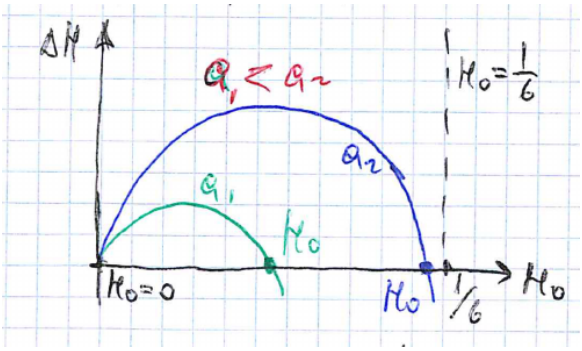
$$\begin{aligned}\Delta H(\Gamma_\varepsilon) &= \varepsilon M(H_0) + O(\varepsilon^2),\\ M(H_0) &= \oint_{\Gamma_0} y(a-x)\,dx\\ &= 2\int_{x_{\min}}^{x_{\max}} \left(2H_0 - x^2 + \frac{2}{3}x^3\right)^{1/2} (a-x)\,dx.\end{aligned}$$

Here $x_{\min} < 0 < x_{\max} < 1$ are the two turning points of the bounded orbit, satisfying $H_0 = H(x,0)$. The integral is generally evaluated numerically. The lecture reports no surviving periodic orbit for $a \le 0$ or $a > 1/7$, and a surviving periodic orbit for $0 < a < 1/7$, in this small-perturbation analysis. The leading energy change increases with a, since

$$\frac{\partial M}{\partial a} = 2\int_{x_{\min}}^{x_{\max}} \left(2H_0 - x^2 + \frac{2}{3}x^3\right)^{1/2} dx > 0.$$



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Source: Lecture12.pdf, p. 3 (original crop).

The leading-order homoclinic threshold. As $H_0 \uparrow 1/6$, the periodic orbit approaches the homoclinic loop, with $x_{\min} \to -1/2$ and $x_{\max} \to 1$. On its upper branch,

$$\left(\frac{1}{3} - x^2 + \frac{2}{3}x^3\right)^{1/2} = \frac{1}{\sqrt{3}}(1-x)\sqrt{1+2x}.$$

Thus the limiting energy balance can be evaluated exactly:

$$\begin{aligned}M(1/6) &= \frac{2}{\sqrt{3}}\int_{-1/2}^{1} (1-x)\sqrt{1+2x}(a-x)\,dx\\ &= \frac{6}{5}\left(a - \frac{1}{7}\right).\end{aligned}$$

This gives $a = 1/7$ as the leading-order threshold. Higher-order terms may shift the threshold for a fixed nonzero ε; also, the regular finite-period expansion above is not uniform at the separatrix.

4.5 Stability of periodic orbits: Floquet analysis

Consider an autonomous flow $\dot{\bar{x}} = \bar{f}(\bar{x})$ in $\mathbb{R}^n$, with a periodic orbit $\bar{x} = \bar{X}(t)$ of period T. Write a nearby solution as

$$\bar{x}(t) = \bar{X}(t) + \bar{\eta}(t).$$

Taylor expansion gives

$$\dot{\bar{X}} + \dot{\bar{\eta}} = \bar{f}(\bar{X} + \bar{\eta}) = \bar{f}(\bar{X}) + A(t)\bar{\eta} + O(\|\bar{\eta}\|^2), \qquad A_{ij}(t) = \left.\frac{\partial f_i}{\partial x_j}\right|_{\bar{X}(t)}.$$

In the linear limit,

$$\dot{\bar{\eta}} = A(t)\bar{\eta}, \qquad A(t+T) = A(t).$$

This is a linear system with periodic coefficients.

1. Define the evolution matrix $\Phi(t)$

By linearity, the map from the initial perturbation to its value at time t is represented by a matrix:

$$\Phi(t) : \bar{\eta}(0) \longmapsto \bar{\eta}(t), \qquad \boxed{\bar{\eta}(t) = \Phi(t)\bar{\eta}(0)}.$$

In components, following the lecture's notation,

$$\eta_i(t) = \Phi_{ij}(t)\eta_j(0), \qquad \Phi_{ij}(0) = \delta_{ij} \quad \Longleftrightarrow \quad \Phi(0) = I.$$

Repeated indices are summed. The jth column of $\Phi(t)$ is the solution starting from the jth unit vector: it records how that initial direction of perturbation evolves.

Substitute into $\dot{\eta}_i = A_{ik}(t)\eta_k$:

$$\dot{\Phi}_{ij}(t)\eta_j(0) = A_{ik}(t)\Phi_{kj}(t)\eta_j(0), \qquad \text{for every } \bar{\eta}(0).$$

Thus

$$\dot{\Phi}_{ij} = A_{ik}(t)\Phi_{kj}, \qquad \boxed{\dot{\Phi}(t) = A(t)\Phi(t), \quad \Phi(0) = I}.$$

This initial-value problem determines $\Phi(t)$; no exponential formula is needed to define it. The component equations are coupled through the sum over k.

2. Define the one-period matrix B

Evaluate the evolution matrix after one period:

$$\boxed{B := \Phi(T)}, \qquad \bar{\eta}(T) = B\bar{\eta}(0).$$

This is the monodromy matrix. Unlike the time-dependent $\Phi(t)$, B is one fixed matrix, describing evolution over a full period. Periodicity of A and uniqueness imply

$$\Phi(t+T) = \Phi(t)B.$$

Indeed, both sides solve the same matrix differential equation and equal B at $t = 0$. Consequently,

$$\Phi(T) = B, \qquad \Phi(2T) = B^2, \qquad \boxed{\Phi(mT) = B^m \quad (m = 0, 1, 2, \ldots)}.$$

Hence $\bar{\eta}(mT) = B^m\bar{\eta}(0)$: repeated multiplication by B determines the evolution over successive periods. These conclusions do not require matrices $A(s)$ and $A(t)$ to commute.

3. Why does the lecture introduce an integral?

Having defined Φ and B, the next question is how to compute them. For a scalar equation,

$$\dot{u} = a(t)u \quad \Longrightarrow \quad u(t) = \exp\left(\int_0^t a(s)\,ds\right) u(0).$$

The lecture attempts the analogous step for the matrix equation $\dot{\Phi} = A(t)\Phi$.

Correction to the lecturer's original formula

Lecture12.pdf, p. 4, writes, immediately after "integrate, find",

$$\Phi_{ij}(t) = B_{ik}(t)\Phi_{kj}(0), \qquad \widehat{B}(t) = \exp\left(\int_0^t \widehat{A}(t')\,dt'\right),$$

with the ordinary matrix exponential defined by

$$e^{\widehat{C}} = \sum_{k=0}^{\infty} \frac{\widehat{C}^k}{k!}.$$

Here hats denote matrices. Since $\Phi(0) = I$, the lecture's time-dependent $\widehat{B}(t)$ must be $\Phi(t)$, and its constant $\widehat{B} = \widehat{B}(T)$ is our monodromy matrix B.

The issue is the ordinary exponential expression for $\widehat{B}(t)$: it does not follow from the hypotheses given. In general, matrices at different times need not commute. Periodicity $A(t+T) = A(t)$ does not imply commutation, and writing the equation in components does not remove the matrix multiplication or its order.

A sufficient additional condition is

$$A(s)A(t) = A(t)A(s) \qquad \text{for all } s, t.$$

Under this condition the lecture's expression is valid, and

$$\Phi(t) = \exp\left(\int_0^t A(s)\,ds\right), \qquad B = \exp\left(\int_0^T A(s)\,ds\right).$$

For example, if A is constant, $\Phi(t) = e^{tA}$ and $B = e^{TA}$.

In general, the always-valid integral formulation is instead

$$\boxed{\Phi(t) = I + \int_0^t A(s)\Phi(s)\,ds}.$$

Equivalently, one can use a time-ordered exponential,

$$\Phi(t) = \mathcal{T}\exp\left(\int_0^t A(s)\,ds\right),$$

where $\mathcal{T}$ preserves the chronological order of the matrix factors. In practice, solve $\dot{\Phi} = A(t)\Phi$, $\Phi(0) = I$, up to time T and then set $B = \Phi(T)$. The original conclusions $\Phi(T) = B$ and $\Phi(mT) = B^m$ remain valid with this definition of B.

End of lecture 12

Consider an autonomous flow $\dot{\bar{x}} = \bar{f}(\bar{x})$ in $\mathbb{R}^n$, with periodic orbit $\bar{x} = \bar{X}(t)$ of period T. Perturb about this PO, $\bar{x}(t) = \bar{X}(t) + \bar{\eta}(t)$. Then

$$\dot{\bar{X}} + \dot{\bar{\eta}} = \bar{f}(\bar{X} + \bar{\eta}) = \bar{f}(\bar{X}) + \bar{\eta} \cdot \nabla \bar{f}(\bar{X}) + O(\|\bar{\eta}\|^2).$$

In the linear limit,

$$\dot{\bar{\eta}} = \widehat{A}(t)\bar{\eta}, \qquad A_{ij} = \left.\frac{\partial f_i}{\partial x_j}\right|_{\bar{X}(t)}, \qquad A_{ij}(t+T) = A_{ij}(t).$$

This is a set of linear ODEs with periodic coefficients. By linearity, introduce a mapping from $\bar{\eta}(0)$ to $\bar{\eta}(t)$:

$$\eta_i(t) = \Phi_{ij}(t)\eta_j(0), \qquad \Phi_{ij}(0) = \delta_{ij}.$$

Substituting into the linear system, for every $\bar{\eta}(0)$,

$$\dot{\Phi}_{ij}(t)\eta_j(0) = A_{ik}(t)\Phi_{kj}(t)\eta_j(0) \quad \Longrightarrow \quad \dot{\Phi}_{ij} = A_{ik}(t)\Phi_{kj}.$$

Integrate, following the lecturer's matrix-exponential notation:

$$\Phi_{ij}(t) = B_{ik}(t)\Phi_{kj}(0), \qquad \widehat{B}(t) = \exp\left(\int_0^t \widehat{A}(t')\,dt'\right), \qquad e^{\widehat{C}} = \sum_{k=0}^{\infty} \frac{\widehat{C}^k}{k!}.$$

Set $\widehat{B} = \widehat{B}(T)$, $B_{ij} = B_{ij}(T)$. Then

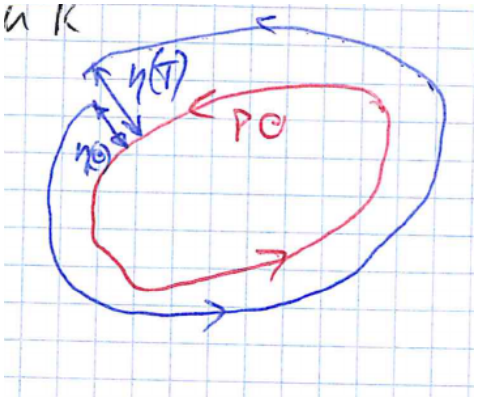
$$\Phi_{ij}(T) = B_{ik}\Phi_{kj}(0) = B_{ik}\delta_{kj} = B_{ij},$$

$$\Phi_{ij}(2T) = B_{ik}\Phi_{kj}(T) = B_{ik}B_{kj} = (\widehat{B}^2)_{ij} = (\widehat{\Phi}(T)^2)_{ij}.$$

At $t = mT$,

$$\Phi_{ij}(mT) = (\widehat{\Phi}(T)^m)_{ij}, \qquad \boxed{\bar{\eta}(mT) = \widehat{\Phi}(T)^m\bar{\eta}(0)}.$$

Changing to a frame in which $\widehat{\Phi}(T)$ is diagonal, whether a perturbation grows or shrinks in an eigendirection is set by its eigenvalue. An eigenvalue with $|\lambda| > 1$ makes the PO unstable. In $\mathbb{R}^2$, part of the perturbation is along the orbit and is passively advected: it neither grows nor shrinks. Thus one eigenvalue is always 1.



Source: Lecture13.pdf, p. 2 (original crop).

Definition The Floquet multipliers of a periodic orbit are the eigenvalues of $\widehat{\Phi}(T)$, denoted λ_i. The Floquet exponents are

$$\mu_i = \frac{1}{T} \ln \lambda_i, \qquad \lambda_i = e^{\mu_i T}.$$

- The PO is hyperbolic if all transverse multipliers satisfy $|\lambda_i| \neq 1$ (apart from the trivial multiplier along the orbit).
- If all transverse $|\lambda_i| < 1$, the PO is asymptotically stable.
- If any $|\lambda_i| > 1$, the PO is unstable.
- In $\mathbb{R}^2$ only one multiplier can be nontrivial, so $\det \widehat{\Phi}(T) = 1 \cdot \lambda = \lambda$.

For a diagonalizable matrix $\widehat{C} = \widehat{Q}\widehat{D}\widehat{Q}^{-1}$, where $\widehat{D} = \mathrm{diag}(d_1, \ldots, d_n)$,

$$\begin{aligned} \det e^{\widehat{C}} &= \det \sum_{k=0}^{\infty} \frac{(\widehat{Q}\widehat{D}\widehat{Q}^{-1})^k}{k!} = \det \left[\widehat{Q} \left(\sum_{k=0}^{\infty} \frac{\widehat{D}^k}{k!} \right) \widehat{Q}^{-1} \right] \\ &= \det \widehat{Q} \, \det e^{\widehat{D}} \, \det \widehat{Q}^{-1} = \prod_{i=1}^{n} e^{d_i} = e^{\sum_i d_i} = e^{\mathrm{tr}\, \widehat{C}}. \end{aligned}$$

The identity $\det e^{\widehat{C}} = e^{\mathrm{tr}\, \widehat{C}}$ also holds for a general square matrix.

Thus, in the lecturer's notation,

$$\det \widehat{\Phi}(t) = \det \exp\left(\int_0^t \widehat{A}(t')\, dt' \right) = \exp\left(\int_0^t \operatorname{tr} \widehat{A}(t')\, dt' \right).$$

In two dimensions, at $t = T$,

$$\lambda = \det \widehat{\Phi}(T) = \exp\left(\int_0^T \operatorname{tr} \widehat{A}(t)\, dt \right).$$

Since $A_{ij} = \partial f_i / \partial x_j$,

$$\operatorname{tr} \widehat{A} = \sum_j \frac{\partial f_j}{\partial x_j} = \nabla \cdot \bar{f}, \qquad \boxed{\lambda = \exp\left(\int_0^T \nabla \cdot \bar{f}(\bar{X}(t))\, dt \right)}.$$

Hence in $\mathbb{R}^2$,

$$\int_0^T \nabla \cdot \bar{f}\, dt \begin{cases} < 0 & \Longrightarrow 0 < \lambda < 1, \quad \text{stable PO}, \\ > 0 & \Longrightarrow \lambda > 1, \quad \text{unstable PO}, \\ = 0 & \Longrightarrow \lambda = 1, \quad \text{non-hyperbolic PO}. \end{cases}$$

In $\mathbb{R}^n$,

$$\exp\left(\int_0^T \nabla \cdot \bar{f}\, dt \right) = \prod_{i=1}^n \lambda_i.$$

A negative integral does not guarantee stability in higher dimensions: one multiplier may still have modulus greater than 1.

Example Determine stability for

$$\dot{r} = r(1 - r^2), \qquad \dot{\theta} = \frac{1}{r},$$

using direct stability analysis and the λ result. The PO is $r = 1$, with $\dot{\theta} = 1$ and $T = 2\pi$. Let $r = 1 + \delta(t)$, $\theta = t + \gamma(t)$. Then

$$\dot{\delta} = (1 + \delta)[1 - (1 + \delta)^2] = -2\delta + O(\delta^2), \qquad \delta(t) = e^{-2t}\delta(0),$$

$$1 + \dot{\gamma} = \frac{1}{1 + \delta} = 1 - \delta + O(\delta^2), \quad \dot{\gamma} = -\delta(0)e^{-2t}, \quad \gamma(t) = \gamma(0) + \tfrac{1}{2}\delta(0)(e^{-2t} - 1).$$

$$\begin{pmatrix}\delta(t)\\ \gamma(t)\end{pmatrix} = \begin{pmatrix} e^{-2t} & 0\\ \frac{1}{2}(e^{-2t}-1) & 1\end{pmatrix}\begin{pmatrix}\delta(0)\\ \gamma(0)\end{pmatrix}, \qquad \widehat{\Phi}(T) = \begin{pmatrix} e^{-2T} & 0\\ \frac{1}{2}(e^{-2T}-1) & 1\end{pmatrix}.$$

Thus $\lambda = \det\widehat{\Phi}(T) = e^{-2T} = e^{-4\pi} < 1$: the PO is stable. Alternatively, in polar coordinates the velocity components are $f_r = \dot{r}$ and $f_\theta = r\dot{\theta}$, so

$$\nabla\cdot\bar{f} = \frac{1}{r}\frac{\partial}{\partial r}(rf_r) + \frac{1}{r}\frac{\partial f_\theta}{\partial\theta} = \frac{1}{r}\frac{\partial}{\partial r}[r^2(1-r^2)] = 2-4r^2.$$

On $r=1$, $\nabla\cdot\bar{f} = -2$, giving $\int_0^T \nabla\cdot\bar{f}\,dt = -4\pi$ and $\lambda = e^{-4\pi}$ again.

4.6 The van der Pol oscillator

One of the foundational dynamical systems. We use it to demonstrate the existence of limit cycles and so-called relaxation oscillations. The van der Pol equation is

$$\boxed{\ddot{x} + \mu(x^2-1)\dot{x} + x = 0}.$$

At small amplitudes there is negative damping, pumping energy into the system. At large amplitudes damping is positive and dissipates energy. Expect the system to live in between, but not at the constant solution $x=1$. This is a special case of a Liénard equation,

$$\ddot{x} + f(x)\dot{x} + g(x) = 0.$$

Use the Liénard transformation

$$y = \dot{x} + F(x), \qquad F(x) = \int_0^x f(s)\,ds,$$

to reduce it to

$$\dot{x} = y - F(x), \qquad \dot{y} = -g(x).$$

For van der Pol,

$$g(x) = x, \qquad F(x) = \mu\left(\frac{x^3}{3} - x\right), \qquad \begin{pmatrix}\dot{x}\\ \dot{y}\end{pmatrix} = \begin{pmatrix} y - \mu(x^3/3 - x)\\ -x\end{pmatrix}.$$

Start by considering two important limits.

(i) $0 < \mu \ll 1$

A near-Hamiltonian system, with

$$H = \tfrac{1}{2}(x^2 + y^2), \quad f_1 = y, \quad f_2 = -x, \quad \varepsilon g_1 = -F(x), \quad \varepsilon g_2 = 0.$$

Without perturbation there is a centre at $(0,0)$. For $\mu \neq 0$, look for a periodic orbit using the energy balance method:

$$\Delta H(H) = \oint \varepsilon g_1 f_1 \, dt.$$

The unperturbed flow is parametrized by

$$x = \sqrt{2H}\sin t, \qquad y = \sqrt{2H}\cos t, \qquad T = 2\pi.$$

Thus, to leading order,

$$\begin{aligned}\Delta H(H) &= \mu \int_0^{2\pi} \left(x^2 - \frac{x^4}{3}\right) dt \\ &= \mu \int_0^{2\pi} \left(2H\sin^2 t - \frac{4}{3}H^2 \sin^4 t\right) dt = 2\pi\mu\left(H - \frac{H^2}{2}\right).\end{aligned}$$

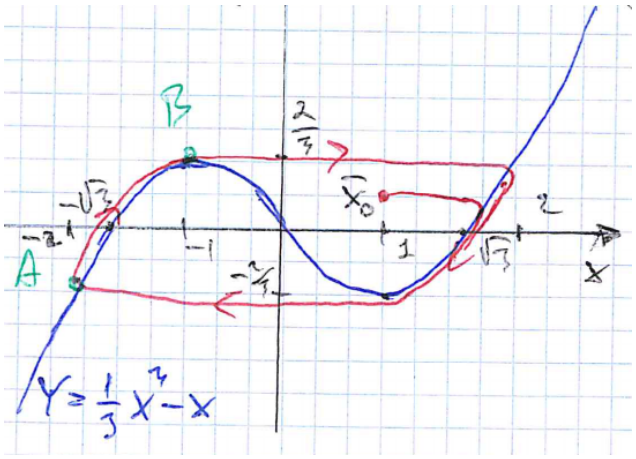
Hence $\Delta H > 0$ for $0 < H < 2$, and $\Delta H < 0$ for $H > 2$. The selected orbit has leading radius $r = 2$, is stable, and has period $P = 2\pi + O(\mu)$.

(ii) $\mu \gg 1$

Rescale $Y = y/\mu$:

$$\begin{pmatrix} \dot{x} \\ \dot{Y} \end{pmatrix} = \begin{pmatrix} \mu[Y - (x^3/3 - x)] \\ -x/\mu \end{pmatrix}.$$

Starting at (x_0, Y_0) of order $(1,1)$ away from $Y = x^3/3 - x$, $|\dot{x}| \gg |\dot{Y}|$. The motion is almost horizontal and fast: an $O(1)$ distance is travelled with speed $O(\mu)$, taking time $O(\mu^{-1})$.



Source: Lecture13.pdf, p. 5 (original crop).

On approaching $Y = x^3/3 - x$ to within $O(\mu^{-2})$, $\dot{x}$ and $\dot{Y}$ become comparable (continued in the next lecture).

End of lecture 13

$$\begin{pmatrix}\dot{x}\\ \dot{y}\end{pmatrix} = \begin{pmatrix} y - \mu(x^3/3 - x) \\ -x \end{pmatrix}.$$

Start by considering two important limits.

(i) $0 < \mu \ll 1$

A near-Hamiltonian system with

$$H = \tfrac{1}{2}(x^2 + y^2), \quad f_1 = y, \quad f_2 = -x, \quad \varepsilon g_1 = -F(x), \quad \varepsilon g_2 = 0.$$

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so, to leading order,

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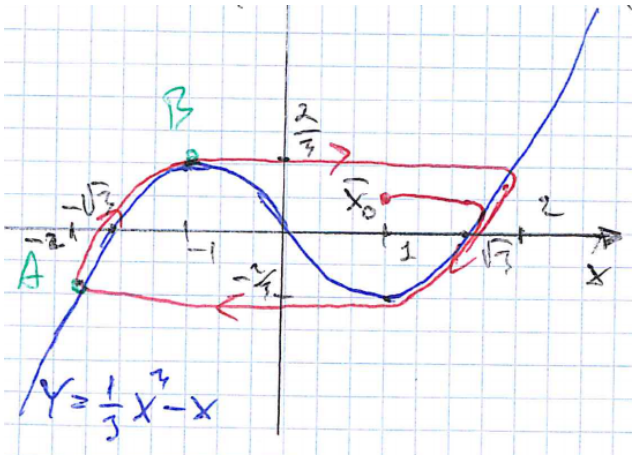
Thus $\Delta H > 0$ for $0 < H < 2$, and $\Delta H < 0$ for $H > 2$. The selected stable orbit has leading radius $r = 2$ and period $P = 2\pi + O(\mu)$.

(ii) $\mu \gg 1$

Rescale $Y = y/\mu$:

$$\begin{pmatrix}\dot{x}\\ \dot{Y}\end{pmatrix} = \begin{pmatrix} \mu[Y - (x^3/3 - x)] \\ -x/\mu \end{pmatrix}.$$

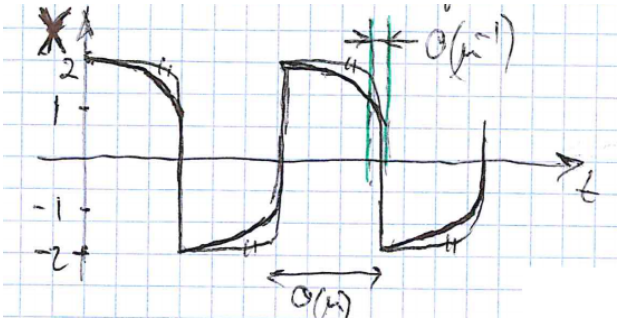
(A) Starting at (x_0, Y_0) of order $(1,1)$ away from $Y = x^3/3 - x$, $|\dot{x}| \gg |\dot{Y}|$: almost horizontal, fast motion, taking time $O(\mu^{-1})$ to travel an $O(1)$ distance at speed $O(\mu)$.



Source: Lecture14.pdf, p. 1 (original crop).

(B) On approaching $Y = x^3/3 - x$ to within $O(\mu^{-2})$, $\dot{x}$ and $\dot{Y}$ become comparable, of order $O(\mu^{-1})$. Travel slowly along the curve until reaching its extremum at $x = \pm 1$. This takes time $O(\mu)$, since the distance is $O(1)$.

(C) After that, quickly move almost horizontally to another branch of the curve, and so on.



Source: Lecture14.pdf, p. 2 (original crop).

This is a relaxation oscillation: slow buildup changing to a fast discharge, repeated periodically. The period is set by the slow buildup phase:

$$\begin{aligned} P &\simeq 2\int_A^B dt = 2\int_{-2/3}^{2/3} \frac{dY}{\dot{Y}} = 2\int_{-2}^{-1} \frac{d(x^3/3 - x)}{-x/\mu} \\ &= 2\mu \int_{-2}^{-1} \frac{1 - x^2}{x}\, dx = \boxed{\mu(3 - 2\ln 2)}. \end{aligned}$$

(iii) Existence of a periodic orbit for every $\mu > 0$

Use Poincaré–Bendixson: show a non-equilibrium forward orbit $O^+(\bar{x}_0)$ remains in a compact set containing no fixed points. The only fixed point is $(0, 0)$, and is unstable. Define

$$H_k(x, y) = \tfrac{1}{2}[x^2 + (y - k)^2] = \tfrac{1}{2} r_k^2,$$

where r_k is the distance to $(0, k)$. Its contours are circles centred at $(0, k)$. Along the flow,

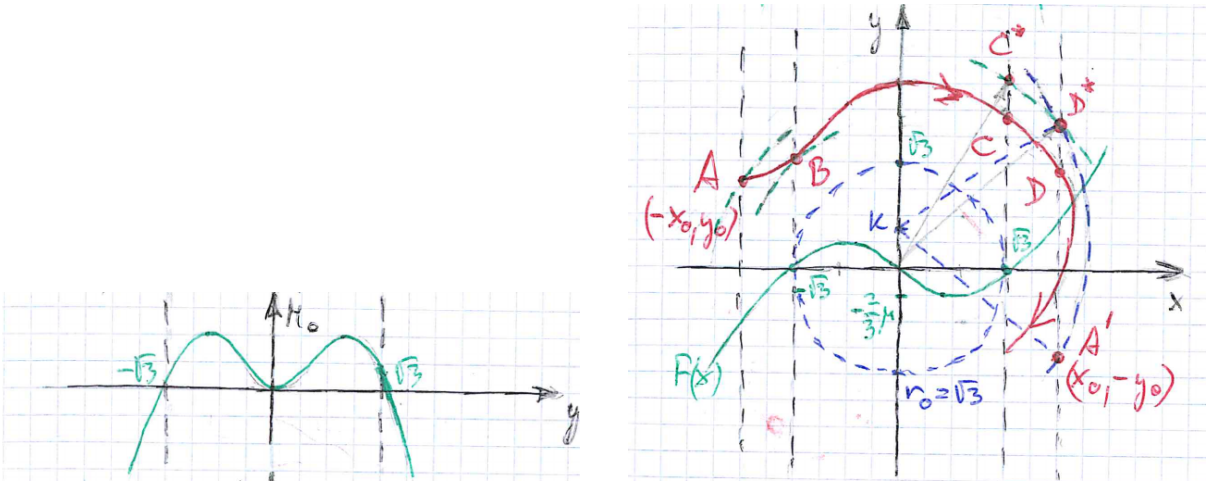
$$\dot{H}_k = \dot{x}\frac{\partial H_k}{\partial x} + \dot{y}\frac{\partial H_k}{\partial y} = kx - xF(x) = kx - \mu x^2 \left(\frac{x^2}{3} - 1\right).$$

In particular, for $k = 0$,

$$\dot{H}_0 = \mu x^2 \left(1 - \frac{x^2}{3}\right) \geq 0 \qquad (|x| < \sqrt{3}),$$

with strict inequality for $x \neq 0$. For any $0 < R_1 < \sqrt{3}$, trajectories on the circle $x^2 + y^2 = R_1^2$ point outward except at $x = 0$, where they are tangent. Thus the disk can only be left in forward time; it provides an inner boundary excluding the fixed point.

Pick $A = (-x_0, y_0)$ and show that $O^+(A)$ cannot cross a suitable closed curve in $\mathbb{R}^2$. Choose $-x_0 < -\sqrt{3}$ and y_0 large enough that $y > F(x)$ for $x < \sqrt{3}$ along the trajectory (see step B).



Source: Lecture14.pdf, p. 3 (original crops).

(A) Move along the orbit to $B = (-\sqrt{3}, y_B)$. Since $\dot{H}_0 < 0$ for $x < -\sqrt{3}$, $r_0(B) < r_0(A)$. But $\dot{y} = -x > 0$ for $x < 0$, so $y_B > y_0$.

(B) Move further to $C = (\sqrt{3}, y_C)$. On BC, $\dot{H}_0 \geq 0$, so r_0 increases. Estimate the change in H_0:

$$\begin{aligned}
\Delta H_0\big|_{B\to C} &= \mu \int_{-\sqrt{3}}^{\sqrt{3}} \left(x^2 - \frac{x^4}{3}\right) \frac{dt}{dx}\, dx \\
&= \mu \int_{-\sqrt{3}}^{\sqrt{3}} \frac{x^2 - x^4/3}{y - F(x)}\, dx \\
&< \frac{\mu}{y_0 - 2\mu/3} \int_{-\sqrt{3}}^{\sqrt{3}} \left(x^2 - \frac{x^4}{3}\right) dx = \frac{4\sqrt{3}\mu}{5(y_0 - 2\mu/3)}.
\end{aligned}$$

Indeed, $\max_{[-\sqrt{3},\sqrt{3}]} F = 2\mu/3$ and

$$\frac{1}{y - F(x)} \leq \frac{1}{y - 2\mu/3} < \frac{1}{y_0 - 2\mu/3}, \qquad y > y_B > y_0 > 2\mu/3.$$

For the positive-x part, $y > y_B$ follows from $H_0(x, y) \geq H_0(B)$ and $x^2 \leq 3$, while y stays positive. Thus

$$H_0(C) = H_0(B) + \Delta H_0\big|_{B\to C} < H_0(A) + \frac{4\sqrt{3}\mu}{5(y_0 - 2\mu/3)} =: H_0(C^*),$$

where

$$\tfrac{1}{2} r_0(C^*)^2 = \tfrac{1}{2}(x_0^2 + y_0^2) + \frac{4\sqrt{3}\mu}{5(y_0 - 2\mu/3)}.$$

Between B and C the orbit lies within this circle.

(C) Continue the orbit to $D = (x_0, y_D)$. On CD, $\dot{H}_0 < 0$, so $r_0(D) < r_0(C)$. Continue the circle $r_0(C^*)$ to $D^* = (x_0, y_{D^*})$ on its upper half. Then $y_{D^*} > y_D$ and

$$\tfrac{1}{2}(x_0^2 + y_{D^*}^2) = \tfrac{1}{2}(x_0^2 + y_0^2) + \frac{4\sqrt{3}\mu}{5(y_0 - 2\mu/3)},$$

so

$$\boxed{y_{D^*}^2 = y_0^2 + \frac{8\sqrt{3}\mu}{5(y_0 - 2\mu/3)}}. \qquad (*)$$

(D) Draw a (blue) circle from D^* to $A' = (x_0, -y_0)$, centred at $(0, k)$, where

$$k = \tfrac{1}{2}(y_{D^*} - y_0) = \frac{4\sqrt{3}\mu}{5(y_0 - 2\mu/3)(y_0 + y_{D^*})}$$
$$< \frac{4\sqrt{3}\mu}{5(y_0 - 2\mu/3)\,2y_0}.$$

On the right-hand circular arc, $x \geq x_0 > \sqrt{3}$. Since $F(x)$ is increasing for $x > 1$, choosing $k < F(x_0)$ gives

$$\dot{H}_k = x[k - F(x)] \leq x[k - F(x_0)] < 0.$$

A sufficient condition is

$$\frac{2\sqrt{3}}{5(y_0 - 2\mu/3)y_0} < x_0 \left(\frac{x_0^2}{3} - 1\right).$$

The parameters can be chosen to satisfy this condition, with y_0 large enough for the preceding orbit segments. Thus the orbit cannot cross the blue circular boundary outward. Together with the orbit segments $ABCD$ and the short joining segment DD^* this gives the upper/right part of a boundary. By the symmetry $(x, y) \mapsto (-x, -y)$, complete it with its reflection, including the reflected arc from A' back towards A. The forward orbit remains inside the resulting bounded region. Exclude an inner circle $r_0 = R_1$, $0 < R_1 < \sqrt{3}$, which the orbit cannot enter in forward time; this excludes the fixed point $(0, 0)$. The resulting compact trapping region contains no fixed points. By Poincaré–Bendixson, a periodic orbit must exist for every $\mu > 0$.

End of lecture 14